

AEEM R402 - Notes

Applied Econometrics I

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Abstract

The notes which were taken from my experience to learn the materials from the course. It is a quick summary of the material which was taken in the course following the literature in the references which are available at the end of the document.

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1 The nature of Regression Analysis

The term regression was coined by Francis Galton.

”regress” of tall parents children’s height to the average height.
”regression to mediocrity”

Regression analysis - study of dependence of a variable upon one or more explanatory variables. With a view to estimating &/or predicting the population mean (average value). Prediction is doing a **statistical relationship**, however strong & however suggestive, can **never** establish **casual connection**.

Different literature refer to the same variables in several ways presented in the table below.

Explained	Explanatory
Depended	Independent
Predicted	Predictor
Regressand	Regressor
Response	Stimulus
Endogenous	Exogenous
Outcome	Covariate
Controlled	Control

Table 1: Different naming combinations used to name the variables.

1.1 Types of Empirical Analysis

There are three main type of Empirical Analysis types

1. Time Series Analysis
2. Cross Section Analysis
3. Pannel Data

Time Series Data

occurring at regular (or irregular) time intervals such as:

1. Secondly
2. Minutely
3. Hourly
4. Daily
5. Weekly
6. Monthly
7. Quarterly
8. Annually
9. Quinquennially (5years)

10. Decimally (10years)

Cross Sectional Data

collection of one or more variables at a certain point in time.

Pooled Data (huge CS)

collection of several cross-sectional databases into a single dataset for analysis.

Panel Data

(Time Series * Cross Sectional) with fixed observants.

Result of the Research is **as good as** the quality of the data.

1.2 Categories of Variables

1.2.1 Ratio Scale

- quantitative measurement system with a:

- true zero
- natural ordering

used for meaningful comparison both in terms of:

- Ratio $\frac{X_1}{X_2}$
- &
- Distance $X_2 - X_1$

1.2.2 Interval Scale

- no true zero
- equal intervals

1.2.3 Ordinal Scale (Qualitative Data)

a scale which has natural ordering

Example: Income class: Upper > Middle > Lower

1.2.4 Nominal Scale

- categorical nature
- no order/rank of inputs
- no quantifiable difference
- no true zero point
- lack of internal consistency

Not suitable for Regression Analysis

2 Two Variable: Regression Analysis

Linear Regression

$$\hat{Y}[E(Y|X_i)] = \beta_1 + \beta_2 * X_i + \varepsilon_i / \nu_i$$

β_1 - intercept

β_2 - slope

ε_i / ν_i - residuals / stochastic disturbance error

3 Two Variable Regress.: Problem of Estimation

3.1 Ordinary Least Square (OLS)

Carl Fredrich Gauss

$$\hat{\nu}_i / \hat{\varepsilon}_i = Y_i - \hat{Y}_i = Y_i - \beta_1 - \beta_2 * X_i$$

3.2 Least Square Criterion

$$\sum \hat{\nu}_i^2 / \hat{\varepsilon}_i^2 = \sum (Y_i - \hat{Y}_i)^2 = \sum (Y_i - \beta_1 - \beta_2 * X_i)^2$$

Optimizing for minimal $\sum \hat{\nu}_i^2 / \hat{\varepsilon}_i^2$

3.3 Normal Equations

$$\beta_2 = \frac{\sum X - \bar{X} * Y - \bar{Y}}{\sum X - \bar{X}} = \frac{\sum X_i * Y_i}{\sum X_i^2}$$

$$\beta_1 = \bar{Y} - \beta_2 * \bar{X}$$

3.4 Assumption of Gaussian (CNLRM)

1. Model parameters are linear.
2. X is non-stochastic. - Regressors are fixed and explained by depended variable.
3. The mean ($\hat{\nu}_i / \hat{\varepsilon}_i$) of ν_i / ε_i is 0.
4. Homoscedasticity - variance of ε is constant or $\text{var}(\varepsilon_i) = \sigma^2$.
5. No autocorrelation between ε_1 & ε_{nth}
. The covariance between first observations and latter one is 0.
Mathematically - $\text{cov}(\varepsilon_1, \varepsilon_{nth} | X) = 0$
6. No covariance between ε_i & X_i .
7. Number of n (observations) > Number of k (parameters).
8. $\text{Var}(X) > 0$.
9. Regression model is correctly specified.
10. There is no perfect multicollinearity.

Formula for	β_2	β_1
Variance	$var\beta_2 = \frac{\hat{\sigma}^2}{\sum X_i^2}$	$var\beta_1 = \frac{\sum X_i^2}{n*\sum X_i^2} * \hat{\sigma}^2$
Standard Error	$se\beta_2 = \frac{\hat{\sigma}}{\sqrt{\sum X_i^2}}$	$se\beta_1 = \sqrt{\frac{\sum X_i^2}{n*\sum X_i^2}} * \hat{\sigma}$

Table 2: Variance & Standard error formulas for each coefficient - $\hat{\sigma}$ - model

3.5 Formulas regarding the coefficient

Last subsection dedicated to crucial formulas about Regression Analysis

Model Variance defined through Residual Sum of Squares (RSS)

$$\hat{\sigma}^2 = \frac{\sum \hat{U}_i^2}{n-2}$$

The concept of Sums of Estimates & Residuals

$$TSS = \sum_{i=1}^n (Y_i - \hat{Y})^2$$

$$ESS = \sum_{i=1}^n (\bar{Y} - \hat{Y})^2$$

$$RSS = \sum_{i=1}^n (Y_i - \bar{Y})^2$$

$$TSS = ESS + RSS$$

degrees of freedom

$$(n-1) = (n-2) + 1$$

The Coefficient of Determination

$$R^2 = 1 - \frac{RSS}{TSS} = \frac{ESS}{TSS}$$

4 Classical Normal Linear Regression (CNLRM)

4.1 The normality Assumption for ν_i/ε_i

Mean of the Residuals is 0 or where close to 0.

$$E(\bar{\nu}_i/\bar{\varepsilon}_i) = 0$$

Variance of the Residuals is constant. (Homoscedasticity)

$$E[\nu_i/\varepsilon_i - \bar{\nu}_i/\bar{\varepsilon}_i]^2 = E(\nu_i^2/\varepsilon_i^2) = \sigma^2$$

Covariance of the first and n-th Residuals is 0. (No autocorrelation)

$$\text{Cov}(\varepsilon_1, \varepsilon_{nth}) = [(\varepsilon_1 - \bar{\varepsilon}_1) * (\varepsilon_{nth} - \bar{\varepsilon}_{nth})] = 0$$

Residuals are Normally & Independently Distributed

$$\varepsilon_i \sim N(0, \sigma^2)$$

4.2 Estimations about OLS under the Normality Assumptions

1. ε_i is unbiased.
2. ε_i has minimal variance.
3. ε_i & outliers are mutually consistent.
4. $\hat{\beta}_1 \sim N(\beta_1, \sigma_{\hat{\beta}_1}^2)$
 $Z \sim N(0, 1)$
5. $\hat{\beta}_2 \sim N(\beta_2, \sigma_{\hat{\beta}_2}^2)$
 $Z \sim N(0, 2)$
6. $(n-2) * (\hat{\sigma}_{\varepsilon}^2) \sim \chi^2 * (n-2)$
7. $(\hat{\beta}_1, \hat{\beta}_2) \propto \hat{\sigma}^2$
8. Best linear unbiased estimators (BLUE) = U_i + Best unbiased estimators (BUE).

To sum up $Y_i \sim N(\beta_1 + \beta_2 * X_i, \sigma^2)$

5 Interval Estimation & Hypothesis Testing

5.1 Confidence limits

$$(\beta_2 - \delta \leq \beta_2 \leq \beta_2 + \delta)$$

$$\text{Lower lim} \leq \beta_2 \leq \text{Upper lim}$$

$$\text{Probability} = 1 - \alpha \text{ (significance level)}$$

$$\delta = z * \beta$$

z - critical value from the normal distribution for the selected α .

5.2 Confidence Intervals

the range where to expect the parameters to lay.

T distribution based

$$\hat{\beta}_2 - t_{\alpha/2} \cdot se(\hat{\beta}_2) \leq \hat{\beta}_2 \leq \hat{\beta}_2 + t_{\alpha/2} \cdot se(\hat{\beta}_2)$$

$t_{\alpha/2}$ - is the critical value from the t-distribution.
Based on the σ^2 being distributed under χ^2

$$\text{Probability of } (\chi^2 - \alpha/2 \leq \chi^2 \leq \chi_{\alpha/2}) = 1 - \alpha$$

$$\text{Probability of } \left(\frac{(n-2) * \sigma}{\chi^2_{1-\alpha/2}} \leq \sigma^2 \leq \frac{(n-2) * \sigma}{\chi^2_{\alpha/2}} \right) = 1 - \alpha$$

5.3 Tails

There are one (directional) & two sided tails.

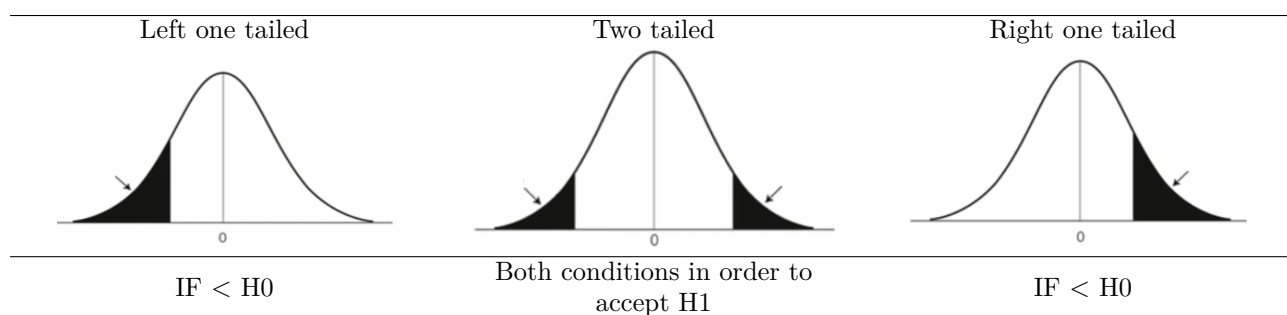


Table 3: Illustration of the tails and H0 | Source: sophia.org

5.4 Choosing α (Significance level)

The level of significance (α) represents the probability of committing Type I error. Usually set in economics at 5% $\rightarrow 0.95$ (1 in 20 is a false positive). Other used levels are 1% $\rightarrow 0.99$ (conservative or 1 in 100), and 10% $\rightarrow 0.9$ (explanatory or 1 in 10).

There is a trade-off effect between α (Type I error) nad β (Type II error)

Higher $\alpha \rightarrow$ Lower β

Type I error $\downarrow$ = Type II error $\uparrow$

Lower $\alpha \leftarrow$ Higher β

Type I error $\uparrow$ = Type II error $\downarrow$

Type I - False positive = Type II - False negative

5.5 P values

The p-value is the probability of observing data at least as extreme as the sample result, assuming the null hypothesis is true. If the p-value is less than the chosen significance level , we reject the null hypothesis.

There are t-test & z-test tables for P-values at set α & df (degrees of freedom)

5.6 Statistical Significance

If the p-value is below the chosen significance level (or the confidence interval excludes the null value), we conclude the result is statistically significant — meaning the observed effect is unlikely to be due to sampling variability alone.

5.7 Economic Significance

or practical significance refers to the magnitude and real-world importance of an effect — whether the size of the estimated effect matters in policy or monetary terms, not just whether it is statistically different from zero.

5.8 F-ratio

The F-ratio (F-statistic) is the test statistic used to compare explained variance to unexplained variance in a linear model.

$$F = \frac{\frac{ESS}{1}}{\frac{RSS}{n-2}} = \frac{MSS_{ESS}}{MSS_{RSS}}$$

$$F = \frac{\hat{\beta}_1^2}{\frac{\hat{\sigma}^2}{\sum_i (X_i - \bar{X})^2}}, \quad \hat{\sigma}^2 = \frac{RSS}{n-2}$$

$$F(1, n-2) = t_{n-2}^2$$

5.9 Normality Tests

5.9.1 Histogram of Residuals

- Resemblance of the Normal Distribution
- Mean near 0.
- Skewness 0.

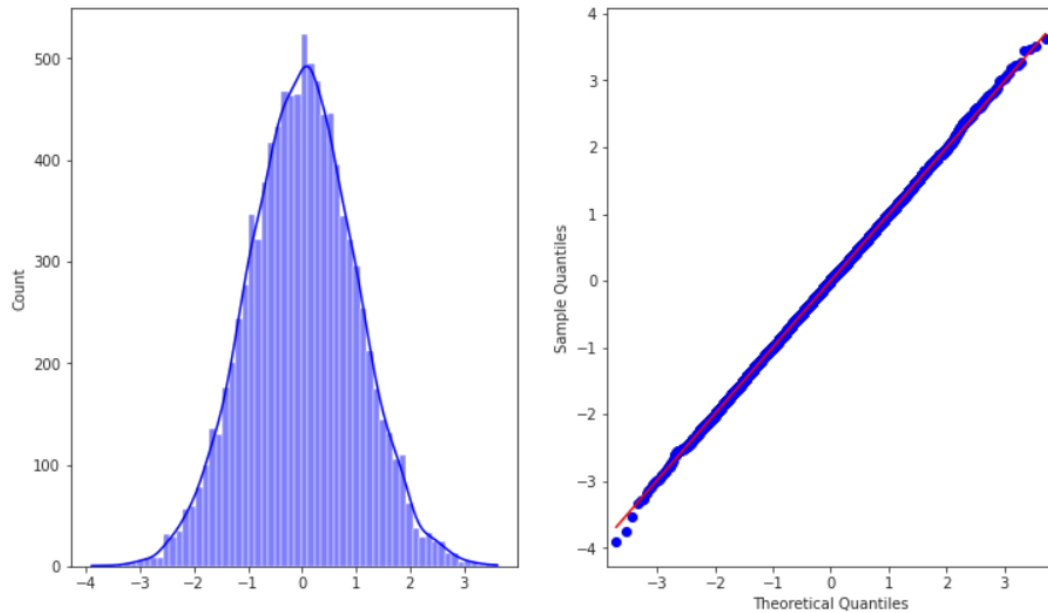
- Kurtosis 3.
- Variance consistent across all levels.

$$\text{Skewness} = \frac{n}{(n-1)(n-2)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{\sigma} \right)^3$$

$$\text{Kurtosis} = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{\sigma} \right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)}$$

5.9.2 Normal Probability Plot (NPP)

Shape analysis Normal distribution plot over Residual Plot. A value off the 45° line is the outlier from the normal distribution.



Source:

[geeksforgeeks.org](https://www.geeksforgeeks.org/)

5.9.3 Jarque-Bera (JB) Test

If $JB = 0$ then we have a normal distribution of residuals.

$$JB = \frac{n}{6} \left(S^2 + \frac{(K-3)^2}{4} \right)$$

n = number of observations

S = skewness

K = sample kurtosis

6 Extensions of the Linear Regression Model

Doing the regression through the origin, assumes the intercept (β_1) to be zero.

6.1 OLS Origin

$$\hat{Y}_i = \beta_2 \cdot X_i + \nu_i / \varepsilon_i \quad , \quad \beta_2 = \frac{\sum Y_i \cdot X_i}{\sum X_i^2}$$

6.2 Special Properties

Residual's mean $\neq 0$. R^2 could be minus because intercept is absent.

$$R^2 = \frac{\sum X_i * Y_i}{\sqrt{\sum X_i^2 * \sum Y_i^2}}$$

6.3 Standardizing Regression Models

The standardization of variables is performed as follows:

1. Subtract the mean: $X_i - \bar{X}$
2. Divide by the standard deviation: $\frac{X_i - \bar{X}}{\sigma}$

Standardized vars have a mean($\bar{X}$) of 0 and a standard deviation(σ) of 1.

Standardized linear regression model formula

$$Y^* = \beta_2^* * X^* + \nu_i^* / \varepsilon_i^*$$

$$Y^* = \frac{Y - \bar{Y}}{\sigma(Y)}$$

$$X^* = \frac{X - \bar{X}}{\sigma(X)}$$

$$\beta_2^* = \beta_2 * \frac{\sigma(Y)}{\sigma(X)}$$

6.4 Functional Forms of Models

Transformation to log or other nonlinear forms is often used to reduce heteroskedasticity and to mitigate skewness in the distribution of residuals.

Main functional forms are:

1. Double log models express both the dependent and independent variables in logarithmic form, allowing for elasticity interpretation. They are often used in economic modeling to analyze proportional relationships in growth rates.
2. Semi-log models (log-linear & linear-log) relate one variable in logarithmic form while keeping the other linear. They are commonly used for handling exponential growth and analyzing percentage changes, particularly in fields like economics and finance.
3. Reciprocal forms express one variable as a function of the reciprocal of another, suitable for cases where the effects diminish at higher levels, such as in modeling diminishing returns on investment or analyzing relationships that reverse at higher input levels.

6.4.1 Log Log (Double Log)

Original (non-linearized)

$$Y_i = \beta_1 + X_i^{\beta_2} + e^{\nu_i/\varepsilon_i}$$

Linearized version

$$\ln(Y) = \ln(\beta_1) + \beta_2 * \ln(X_i) + \nu_i/\varepsilon_i$$

Elasticity Interpretation

$$\beta_2 = \frac{\% \Delta Y}{\% \Delta X}$$

6.4.2 Log Lin

Original (non-linearized)

$$Y_i = \beta_1 * e^{\beta_2 * X_i} * \nu_i/\varepsilon_i$$

Linearized version

$$\ln(Y_i) = \ln(\beta_1) + \beta_2 * X_i + \ln(\nu_i/\varepsilon_i)$$

Elasticity Interpretation

$$\% \Delta Y = (e^{\beta_2} - 1) * 100 \quad , \approx \beta_2 * 100 \quad \text{only for small } \beta_2$$

6.4.3 Growth rate analysis using Log Lin

Original (non-linearized)

$$Y_t = Y_0 * (1 + r)^t + \nu_t/\varepsilon_t$$

Linearized version

$$\ln(Y_t) = \ln(Y_0) + t * \ln((1 + r)) + \nu_t/\varepsilon_t$$

Elasticity Interpretation

$$\% \text{growth} = (e^{\beta_2} - 1) * 100 \quad , \approx \beta_2 * 100 \quad \text{only for small } \beta_2$$

6.4.4 Lin Log

Original (Linearized)

$$Y_i = \beta_1 + \beta_2 * \ln X_i + \nu_i/\varepsilon_i$$

Definition of β_2

$$\beta_2 = \frac{\Delta Y}{\Delta \ln(X)}$$

Elasticity Interpretation approximation only for small β_2

$$\Delta Y \approx \beta_2 * \frac{\Delta X}{X}$$

6.4.5 Reciprocal Models

Original

$$Y_i = \beta_1 + \beta_2 * \frac{1}{X_i} + \nu_i/\varepsilon_i$$

Definition of β_2

$$\beta_2 = \frac{\Delta Y}{\Delta \frac{1}{X}}$$

Small change approximation

$$\Delta Y \approx -\beta_2 * \frac{\Delta X}{X^2}$$

6.4.6 Log Reciprocal Models

Original

$$\ln(Y_i) = \beta_1 + \beta_2 * \frac{1}{X_i} + \nu_i/\varepsilon_i$$

Definition of β_2

$$\beta_2 = \frac{\Delta \ln(Y)}{\Delta \frac{1}{X}}$$

Small change approximation

$$\Delta \ln(Y) \approx -\beta_2 * \frac{\Delta X}{X^2}$$

Elasticity Interpretation approximation only for small β_2

$$\% \Delta Y \approx -100 * \beta_2 * \frac{\Delta X}{X^2}.$$

7 Extended model: Estimation Problems

Two or more regressor is a common practice.

$$Y_i = \beta_1 + \beta_2 * X_{2i} + \beta_3 * X_{3i} + \nu_i / \varepsilon_i$$

7.1 Follows normal formulas of $\hat{\beta}_1$, $\hat{\beta}_2$ & $\hat{\beta}_3$ and model variance.

$$\hat{\beta}_1 = \bar{Y} - \hat{\beta}_2 * \bar{X}_2 - \hat{\beta}_3 * \bar{X}_3$$

$$\hat{\beta}_2 = \frac{(\sum Y_i * X_{2i}) * \sum X_{3i}^2 - (\sum Y_i * X_{3i}) * \sum X_{2i} * X_{3i}}{\sum X_{2i}^2 * \sum X_{3i}^2 - (\sum X_{2i} * X_{3i})^2}$$

$$\hat{\beta}_3 = \frac{(\sum Y_i * X_{3i}) (\sum X_{2i}^2) - (\sum Y_i * X_{2i}) * (\sum X_{2i} * X_{3i})}{(\sum X_{3i}^2) * (\sum X_{2i}^2) - (\sum X_{2i} * X_{3i})^2}$$

$$\sigma^2 = \frac{\sum \hat{\nu}_i^2 / \varepsilon_i^2}{n - 3 - df}$$

7.2 Variance, Standard Error, Covariance for each coefficient or where applicable.

$$\text{var}(\hat{\beta}_1) = \frac{1}{n} * \frac{\bar{X}_2^2 * \sum X_{3i}^2 + \bar{X}_3^2 * \sum X_{2i}^2 - 2 * \bar{X}_2 * \bar{X}_3 * \sum X_{2i} * X_{3i}}{\sum X_{2i}^2 * \sum X_{3i}^2 - (\sum X_{2i} * X_{3i})^2} * \sigma_2$$

$$\text{se}(\hat{\beta}_1) = +\sqrt{\text{var}(\hat{\beta}_1)}$$

$$\text{var}(\hat{\beta}_2) = \frac{\sum X_{3i}^2}{\sum X_{2i}^2 * \sum X_{3i}^2 - (\sum X_{2i} * X_{3i})^2} * \sigma_2$$

$$\text{var}(\hat{\beta}_2) = \frac{\sigma_2}{\sum X_{2i}^2 * (1 - r_{23}^2)}$$

$$\text{se}(\hat{\beta}_2) = +\sqrt{\text{var}(\hat{\beta}_2)}$$

$$\text{var}(\hat{\beta}_3) = \frac{\sigma_2}{\sum X_{3i}^2 * (1 - r_{23}^2)}$$

$$\text{var}(\hat{\beta}_3) = \frac{\sum X_{2i}^2}{\sum X_{2i}^2 * \sum X_{3i}^2 - (\sum X_{2i} * X_{3i})^2} * \sigma_2$$

$$\text{se}(\hat{\beta}_3) = +\sqrt{\text{var}(\hat{\beta}_3)}$$

$$\text{cov}(\hat{\beta}_2, \hat{\beta}_3) = \frac{-r_{23} * \sigma^2}{(1 - r_{23}^2) * \sqrt{\sum X_{2i}^2} * \sqrt{\sum X_{3i}^2}}$$

$$r_{23}^2 = \frac{(\sum X_{2i} * X_{3i})^2}{\sum X_{2i}^2 * \sum X_{3i}^2}$$

7.3 R^2 , $\bar{R}^2$ and Modified R^2 in a multiple coefficient model

$$R^2 = \frac{\hat{\beta}_2 * \sum Y_i * X_{2i} + \hat{\beta}_3 * \sum Y_i * X_{3i}}{\sum Y_i^2} = \frac{ESS}{TSS}$$

$$\bar{R}^2 = 1 - \frac{\frac{\sum \hat{\nu}_i^2 / \varepsilon_i^2}{n-k}}{\frac{\sum Y_i^2}{n-1}} = 1 - \frac{\sigma_{\nu_i/\varepsilon}^2}{\sigma_{model}^2}$$

$$\text{Modified } R^2 = \frac{1-k}{n} * R^2$$

Maximizing R^2 is **NOT** the goal but rather
removing **Econometric** Problems & getting **Practical** Significance.

7.4 Cobb Douglass Function

Original

$$Y_i = \beta_1 * X_{2i}^{\beta_2} * X_{3i}^{\beta_3} * e^{\nu_i/\varepsilon_i}$$

Y - output

X_2 - labor input

X_3 - capital input

Linearized

$$\ln(Y_i) = \ln(\beta_1) + \beta_2 * \ln(X_{2i}) + \beta_3 * \ln(X_{3i}) + \nu_i/\varepsilon_i$$

Scale Interpretation

$$\begin{array}{ll} \beta_2 + \beta_3 \leq 1 \approx & 2x \text{ Input} = \downarrow_2 \text{ Output} \\ \beta_2 + \beta_3 = 1 \approx & 2x \text{ Input} = 2x \text{ Output} \\ \beta_2 + \beta_3 \geq 1 \approx & 2x \text{ Input} = \uparrow^2 \text{ Output} \end{array}$$

8 Estimation Problems: T-test & F-test

8.1 T-test

$$t = \frac{\hat{\beta}_n - \beta_n (\approx 0)}{\sigma_{(\hat{\beta}_n)}} = \frac{\hat{\beta}_n}{\sigma_{(\hat{\beta}_n)}}$$

Higher is better, if $|t| > t_{\text{critical for } \alpha}$ H_0 is rejected.

8.2 F-test

$$F = \frac{\frac{ESS}{k-1}}{\frac{RSS}{n-k}}$$

Higher is better, more of the variance of the model is explained by ESS.

8.3 Restricted Least Squares (RLS)

is a statistical technique used in regression analysis to estimate the parameters of a model while imposing certain constraints on them. Unlike ordinary least squares (OLS), which estimates parameters freely, RLS allows for specific restrictions, enabling researchers to test hypotheses or incorporate prior knowledge about the parameter values. Example both exogen coefficients have a sum equal to 1:

$$\beta_2 = 1 - \beta_3 ; \quad \beta_3 = 1 - \beta_2$$

Implementation in Cobb-Douglas form:

$$\ln(Y_i) = \beta_0 + (1 - \beta_3) * \ln(X_{2i}) + \beta_3 * \ln(X_{3i}) + \nu_i / \varepsilon_i$$

Restricted form of it:

$$\ln(Y_i - \ln(X_{2i})) = \beta_0 + \beta_3 * (\ln(X_{3i}) - \ln(X_{2i})) + \nu_i / \varepsilon_i$$

And in a multiplicative form of it:

$$\ln\left(\frac{Y_i}{X_{2i}}\right) = \beta_0 + \beta_3 * \ln\left(\frac{X_{3i}}{X_{2i}}\right) + \nu_i / \varepsilon_i$$

This regression specification of RLS is checking if there is a connection between β_2 & β_3 equal to 1.

8.4 MacKimon White & Davidson (MWD) Test

the test evaluates whether the restricted model significantly differs from the unrestricted model. It helps determine if the constraints imposed on the parameters are valid.

1. Estimate the Linear model
2. Estimate Log-linear Model
3. Obtain Z1
4. Obtain Z2
5. Regress Z2

9 Dummy Variables

artificial binary variables created to present:

- categorical data with two or more categories in format of present (1) & non-present (0).
- threshold of a continuous variable - example if variable $> 100 = 1$, else 0.

9.1 Cautions when using Dummy Variables

9.1.1 Dummy Variable Trap

involving a dummy variable for each category of a qualitative variable along on intercept, which results in perfect multicollinearity.

9.1.2 Choosing the Benchmark Category

is crucial as this category serves as a point of reference for all comparisons. Often, the choice of benchmark affects the interpretation of the coefficients and the results of the analysis.

9.1.3 Interpreting the Intercept

the intercept represents the mean value of the dependent variable for the benchmark category

9.1.4 Coefficient Interpretation

variation of the mean of the category in regard to the benchmark (intercept). Positive coefficient suggests an increase in the mean, whereas a negative coefficient indicates a decrease compared to the benchmark category.

9.1.5 Model Fit & Colinearity

Including multiple dummy variables related to the same underlying categorical variable can lead to collinearity issues, which complicates the estimation of coefficients and can inflate standard errors, ultimately degrading the model's fit.

9.1.6 Sample Size

Having an adequate sample size is vital when including dummy variables in a model. Each category represented by a dummy variable requires sufficient observations to ensure reliable estimation of coefficients. A small sample size can lead to unreliable estimates and may result in overfitting, where the model captures noise instead of the underlying relationship.

9.1.7 Non Linearity

Dummy variables are designed to capture discrete changes in the dependent variable in relation to categorical inputs. However, they may not accurately represent non-linear relationships between these variables and the outcome. It's important to consider whether transformations or interactions might be needed to accurately capture the relationships in cases where non-linearity is suspected.

9.2 Chow Test

statistical method to examine structural stability of regression model overtime or across different groups.

Steps:

1. Fit two models - M_1 & M_2 .

2. Calculate each models RSS & Pooled. - RSS_1 , RSS_2 & RSS_p .

3. Apply the formula.

Formula:

$$F = \frac{\frac{RSS_p - (RSS_1 + RSS_2)}{k}}{\frac{RSS_1 + RSS_2}{n_1 + n_2 - 2 * k}}$$

9.3 Interaction Terms

interaction terms explain joint effect of multiple interacted dummies.

$$\text{example: } Y_i = \beta_1 + \beta_2 * D_2 + \beta_3 * D_3 + \beta_4 * (D_2 * D_3) + \nu_i / \varepsilon_i$$

9.4 Piecewise Linear Regression

models are particularly effective when we suspect that the relationship between the independent and dependent variables is not constant across the entire range of data. Instead, it allows us to define different linear relationships for different segments of the data. The main components and benefits of piecewise regression include:

- **Threshold Level (X^*):** This is the point where the relationship changes, creating two segments of linearity.
- **Different Slopes:** The model captures varying slopes on either side of the threshold, which is critical in understanding how changes in the independent variable affect the dependent variable differently before and after the threshold.
- **Use of Dummy Variables:** To implement this, dummy variables are used effectively to differentiate the segments. For instance, when constructing a regression model, a dummy variable D is created that equals 1 when X is greater than X^* and 0 otherwise.
- **Applications:** This approach is employed in various fields including economics, health sciences, and marketing, particularly where policies or incentives change at specific cut-off points.
- **Model Estimation:** By applying this model, researchers can derive insights into how varying commissions or incentives impact performance metrics differently across sales ranges.

To illustrate yet another use of dummy variables, consider a hypothetical company that remunerates its sales representatives. It pays commissions based on sales in such a manner that up to a certain level, the target, or threshold level X^* , there is one commission structure and beyond that level, another. It is assumed that sales commission increases linearly with sales until the threshold level X^* , after which it continues to increase linearly with sales but at a much steeper rate. Thus, we have a piecewise linear regression consisting of two linear pieces or segments, and the commission function changes its slope at the threshold value. Given the data on commission, sales, and the value of the threshold level X^* , the technique of dummy variables can be used to estimate the differing slopes of the two segments of the piecewise linear regression. We proceed as follows:

$$Y_i = \alpha_1 + \beta_1 X_i + \beta_2 (X_i - X^*) D_i + u_i$$

Where:

- Y_i is the outcome variable (e.g., commission).
- α_1 is the intercept for the first segment.
- β_1 is the slope before the threshold.
- β_2 modifies the slope after the threshold.

10 Multicollinearity

10.1 Sources of Multicollinearity

10.1.1 Sample Phenomenon

Multicollinearity can arise due to the specific characteristics of the sample. If the sample lacks diversity and has repeated or similar values for one or more independent variables, it can lead to inflated correlations between those variables.

10.1.2 Data Collection

The methods used to collect data can introduce multicollinearity. For instance, if data is collected through surveys or instruments that measure similar constructs, the resulting variables may become highly correlated.

10.1.3 Overdetermination of the Model

Including too many predictors in a model relative to the number of observations can cause overdetermination. This situation often results in multicollinearity, as the added variables may overlap in the information they provide.

10.1.4 Constrains on the Population of the Sample

If the sample is too narrowly defined (for example, focusing on a specific demographic), it can lead to multicollinearity. The limited diversity among variables can exaggerate correlations between them.

10.1.5 Common Trend

When independent variables share a common trend over time, such as seasonality or economic changes, they may become correlated. This shared trend can lead to multicollinearity, making it difficult to distinguish the individual effects of those variables.

10.1.6 Model Specification

Improper model specification, including omitted variable bias or inappropriate functional forms, can lead to multicollinearity. If important variables are missing or if the model does not correctly represent the relationships between variables, it can inflate correlation among those included.

10.1.7 Micronumerosity

This term refers to having an insufficient number of observations relative to the number of predictors. When the sample size is small, even modest correlations between independent variables can result in multicollinearity due to the lack of data to support the relationships.

10.2 Consequences of Multicollinearity

10.2.1 Large Variances

When multicollinearity is present, the variances of the coefficient estimates can become inflated. This leads to larger standard errors, making it more difficult to determine the statistical significance of the predictors. High variances can indicate that changes in one predictor may lead to extreme changes in estimates for other predictors.

10.2.2 Wider Confidence Intervals

Due to inflated variances, the confidence intervals for the coefficient estimates widen. This lack of precision means that the true value of the coefficients may lie far from the estimates provided, complicating decision-making and interpretations of the results.

10.2.3 Insignificant T-Ratio

The T-ratio, which is calculated as the coefficient divided by its standard error, can become insignificant in cases of multicollinearity. This means that, even if a variable appears to be relevant, it may not show statistical significance due to the inflated standard errors stemming from multicollinearity.

10.2.4 High R^2 with T-Insignificant

It's possible to encounter a situation where the model has a high R^2 value, indicating that a large proportion of the variance in the dependent variable is explained by the model, yet many of the individual T-ratios are insignificant. This inconsistency can create confusion, as it suggests a good fit in aggregate while failing to substantiate the predictive power of individual predictors.

10.2.5 Sensitivity to Data Changes

Models exhibiting multicollinearity tend to be overly sensitive to changes in the data. Small modifications in the datasets, such as adding new observations or removing outliers, can lead to significant fluctuations in coefficient estimates. This sensitivity compromises model stability and reliability, raising concerns about its generalizability to other datasets.

10.3 Detection of Multicollinearity

10.3.1 High R^2

A high R^2 value in regression indicates that a large proportion of the variance in the dependent variable is explained by the independent variables. However, when multicollinearity is present, R^2 may still be misleadingly high. This is because correlated predictors might inflate the R^2 without contributing unique explanatory power to the model. Therefore, while a high R^2 suggests a good fit, it does not guarantee that the model will provide reliable coefficient estimates or interpretations. Assessing R^2 alongside other diagnostic tools is essential for accurate analysis.

10.3.2 High Pair Wise Correlations

High pairwise correlations among independent variables are a strong indicator of multicollinearity. When two or more predictors exhibit significant correlation, it becomes challenging to determine their individual effects on the dependent variable. Correlated variables can create overlapping information, which complicates the statistical interpretation of the regression coefficients. Investigating the correlation matrix allows researchers to identify potentially problematic pairs of variables that might require further scrutiny or removal from the model. A correlation coefficient above 0.7 is typically a flag for concern.

10.3.3 Auxiliary Regressions

Auxiliary regressions serve as a powerful method to diagnose multicollinearity. In this approach, one independent variable is regressed on the other independent variables, and the R^2 of this auxiliary regression is examined. A high R^2 value in the auxiliary regression suggests that the variable in question is highly explainable by the other predictors, indicating multicollinearity. This method provides insight into which variables are contributing to redundant information, helping researchers make informed decisions about variable inclusion or exclusion in the main model.

10.3.4 Eigen Values & Condition Index (CI)

Eigenvalues and the Condition Index (CI) are mathematical tools used to assess multicollinearity. Eigenvalues provide information about the variance associated with each dimension in the predictor space, while the CI indicates how much the variance of the coefficient estimates may be inflated due to multicollinearity. Typically, a CI greater than 30 is considered a sign of severe multicollinearity. Evaluating these metrics allows researchers to quantify multicollinearity and to understand its potential impact on the stability and reliability of regression estimates.

10.3.5 Variation Inflating Factor (VIF)

The Variance Inflation Factor (VIF) quantifies the extent of multicollinearity in regression analysis. It measures how much the variance of an estimated regression coefficient increases due to multicollinearity. A VIF value exceeding 10 is commonly considered an indication of significant multicollinearity. By calculating the VIF for each independent variable, researchers can identify which variables contribute most heavily to multicollinearity. This information can guide decisions about whether to remove, combine, or retain certain predictors in the model, ultimately leading to more robust statistical analyses.

10.4 Remedies for Multicollinearity

10.4.1 Do nothing

In some cases, particularly when multicollinearity is not significantly affecting the model outcomes, researchers may choose to take no action. This approach can be viable if the primary interest lies in prediction rather than interpretation of individual coefficients. If the model's overall performance metrics, such as adjusted R^2 and predictive accuracy, remain acceptable, it may be prudent to continue using the current model without modifications. However, careful consideration is necessary to avoid misinterpretation of coefficient estimates.

10.4.2 Making Pooled Data

One effective way to tackle multicollinearity is by pooling data from different sources or groups. By aggregating data sets, researchers can increase variability among the independent variables, reducing the chances of multicollinearity. This approach helps to dilute the correlation effects between predictors, enhancing the model's reliability. However, it's crucial to ensure that the pooled data are conceptually and contextually consistent to avoid introducing bias or misleading results.

10.4.3 Dropping Variables

Dropping one or more highly correlated predictor variables is a straightforward remedy for multicollinearity. This method involves identifying variables that contribute redundant information and removing them from the model. Although this might lead to a reduction in the model's explanatory power, it simplifies interpretation and reduces variance inflation. Care must be taken to evaluate the impact of removing variables on the overall model performance, ensuring that essential predictors aren't eliminated inadvertently.

10.4.4 Transformation of Data

Transforming the data can help manage multicollinearity by changing the scale or distribution of variables. Common techniques include logarithmic, square root, or polynomial transformations. By altering the functional form of certain predictors, researchers can sometimes reduce their correlation with other variables, helping to mitigate multicollinearity. This method can not only improve model fit but may also help in better meeting the assumptions of linear regression.

10.4.5 Adding Additional Data

Collecting and incorporating additional data can also serve as a remedy for multicollinearity. By increasing the sample size, researchers may reduce the correlation between independent variables, particularly if the new data provide a broader context or more variance among predictor values. This approach not only helps combat multicollinearity but also enhances the model's statistical power, allowing for more reliable inference and predictions.

10.4.6 Reducing colinearity with deviations

Using deviations from the mean to create new variables can effectively reduce multicollinearity. By centering the variables (subtracting the mean from each observation), researchers mitigate the correlation among predictors, especially when interactions or polynomial terms are used. This technique helps in isolating the unique contribution of each predictor, improving coefficient estimates without losing valuable information from the original variables.

10.4.7 Factor Analysis

Factor analysis is a multivariate statistical method used to identify underlying relationships among a set of variables. By reducing the number of variables into a smaller set of factors, researchers can alleviate multicollinearity. This technique captures the essence of the original variables while minimizing redundancy. The resulting factors can then be used as predictors in the regression model, leading to a more parsimonious and interpretable outcome.

10.4.8 PCA

Principal Component Analysis (PCA) is a powerful technique for addressing multicollinearity by transforming correlated variables into a set of uncorrelated components. PCA identifies the directions (principal components) that maximize variance in the data, allowing researchers to retain most of the information while reducing dimensionality. By using these components as predictors, multicollinearity is significantly diminished, enabling better model performance and interpretation.

10.4.9 Ridge Regression

Ridge regression is an advanced technique specifically designed to tackle multicollinearity. By adding a penalty term to the loss function, ridge regression shrinks the coefficients of correlated predictors, effectively reducing variance and stabilizing their estimates. This approach allows researchers to retain all variables while minimizing multicollinearity's negative impact. Ridge regression is particularly useful when dealing with datasets where the number of predictors approaches or exceeds the number of observations.

11 Heteroscedacity

11.1 Sources of Heteroscedacity

11.1.1 Cross Sectional Data

Cross-sectional data can often lead to heteroscedasticity due to variations in the characteristics of the units being analyzed. For instance, if a regression model includes individuals from different socioeconomic backgrounds, the variance of the residuals may differ across these groups. This discrepancy can result in non-constant error variances, creating challenges in obtaining valid inference. Analysts must be diligent in assessing and correcting for this variability when using cross-sectional data.

11.1.2 Incorrect Functional Form

An incorrect functional form can lead to heteroscedasticity by failing to capture the true relationship between the independent and dependent variables. If the model misses important nonlinear relationships or interactions, the variance of the residuals may change based on the predicted values. To address this issue, researchers should explore different functional forms (e.g., linear, log-linear, polynomial) and conduct tests to identify the most appropriate specification for the relationship under study.

11.1.3 Incorrect Data Transformation

Applying inappropriate data transformations can introduce heteroscedasticity into a regression model. If transformations are misapplied or insufficient, they may not stabilize the variance of the residuals. For example, failing to logarithmically transform positively skewed data can lead to increasing variance as the predicted values rise. Properly assessing the need for transformation and selecting the right method is critical for ensuring homoscedasticity.

11.1.4 Learn by Doing

The “learn by doing” approach emphasizes that as practitioners gain experience in data analysis, they become more adept at recognizing and addressing issues like heteroscedasticity. Over time, analysts develop a better intuition for detecting patterns in residuals, understanding the nature of their data, and applying appropriate remedies. This practical experience enables researchers to avoid common pitfalls associated with heteroscedasticity and improve their model specifications effectively.

11.1.5 Skewness

Skewness in the distribution of the dependent variable can be a significant source of heteroscedasticity. When the dependent variable exhibits a skewed distribution, the variance of the residuals may vary across different levels of the independent variables. For instance, if the dependent variable is positively skewed, higher values may display greater variability in terms of the error term. Addressing skewness through transformation or different modeling techniques can help stabilize variance.

11.1.6 Higher Incomes

In economic models, higher income levels can lead to heteroscedasticity due to increased variability in consumption or expenditure patterns. Typically, as income rises, individuals may exhibit greater disparity in spending behavior, causing error variance to increase. This increasing variance related to higher incomes can lead to violations of the homoscedasticity assumption in regression models, necessitating careful consideration and potential corrective measures.

11.1.7 Improper Data Collection

Improper or biased data collection methods can introduce heteroscedasticity into the analysis. If data is collected in a manner that systematically overlooks certain groups or contains measurement errors, the variance of the residuals may not remain constant. For example, if a survey only reaches high-income individuals, it may yield results with erratic variance. Ensuring rigorous data collection protocols is vital to minimize the risk of heteroscedasticity.

11.1.8 Omitting an Important Variable

Omitting an important variable from a regression model can lead to spurious relationships and heteroscedasticity. When key predictors are left out, the variance of the error term may be affected, as the model fails to account for variability explained by those omitted variables. This oversight can skew the results and lead to invalid conclusions. Careful variable selection and inclusion processes are essential to mitigate this source of heteroscedasticity.

11.1.9 Presence of Outliers

Outliers can significantly impact the variance of the error terms in regression models, leading to heteroscedasticity. These extreme values may distort the underlying relationships between variables and elevate the variance of residuals, which can create misleading inferences. Identifying and addressing outliers through strategies such as robust regression techniques or transformations is necessary to maintain the validity of the regression analysis.

11.2 Detection of Heteroscedasticity

11.2.1 A priori

A priori methods for detecting heteroscedasticity involve using theoretical knowledge or previous research to anticipate where non-constant variance might occur. Analysts may assume heteroscedasticity based on the nature of the data or underlying processes that suggest variability in error terms. For example, in economic models, the relationship between income and consumption may inherently lead to varying error variances. This informed approach helps analysts focus on specific contexts or variables that are likely to exhibit heteroscedasticity, enabling targeted tests and diagnostic tools to confirm or refute the presence of the issue.

11.2.2 Graphical Methods

Graphical methods are a widely-used approach for visually detecting heteroscedasticity in regression models. By plotting residuals against fitted values or independent variables, analysts can observe patterns in variance. Ideally, the residuals should display a random scatter around zero, indicating homoscedasticity. However, if the plot reveals a systematic pattern, such as a funnel shape, it suggests the presence of heteroscedasticity. Additional plots, like Q-Q plots and scale-location plots, can further confirm non-constant variance. These visual inspections provide immediate insights into the integrity of regression assumptions, guiding subsequent statistical tests and model adjustments.

11.2.3 Park Test

variance test between ν_i^2/ε_i^2 & X_i .

$$\begin{aligned}\sigma_i^2 &= \sigma^2 X_i^\beta \Rightarrow \\ \ln(\sigma_i^2) &= \ln(\sigma^2) + \beta * \ln(X_i) + \nu_i/\varepsilon_i \\ \text{If } \beta &\neq 0 \Rightarrow \text{Heteroscedasticity}\end{aligned}$$

11.2.4 Glejer Test

absolute value test between $\hat{\nu}_i/\hat{\varepsilon}_i = X_i$

$$\text{Direct: } \hat{\nu}_i/\hat{\varepsilon}_i = \beta_1 + \beta_2 * X_i + \nu_i/\varepsilon_i$$

$$\text{Indirect: } \hat{\nu}_i/\hat{\varepsilon}_i = \beta_1 + \frac{\beta_2}{X_i} + \nu_i/\varepsilon_i$$

If $\beta_2 \neq 0 \Rightarrow$ Heteroscedasticity

11.2.5 Goldfeld-Quandt Test

uses variance in the error term across different groups of the explanatory variables.

1. Sort the Dataset: Organize the dataset based on the independent variable X .
2. Divide the Dataset: Split the dataset into two groups based on the independent variable(s), ensuring that the removed middle portion contains at least 10-20% of the entire sample of X .
3. Estimate the Regression Model: Perform ordinary least squares (OLS) regression separately on both groups to obtain residuals.
4. Calculate the Variance of Residuals: Compute the residual sum of squares (RSS) for both groups.
5. Form the Test Statistic: The test statistic is calculated as:

$$\lambda = \frac{RSS_{low}}{RSS_{high}}$$

where RSS_{low} and RSS_{high} are the residual sum of squares for the lower and higher groups, respectively.

6. Decision Rule: If λ follows an F -distribution, we can conclude homoscedasticity if λ is not significantly different from 1. If λ indicates a significant ratio, it suggests heteroscedasticity.

11.2.6 Breusch-Pagan-Godfrey Test

examines the relationship between the squared residuals from a regression model and the independent variables to detect heteroscedasticity. If the variance of the residuals is significantly linked to the independent variables, the test indicates that the assumption of constant variance (homoscedasticity) is violated.

1. Run the original regression model:

$$Y = \beta_0 + \beta_1 * X_1 + \beta_2 * X_2 + \dots + \beta_k * X_k + \nu_i/\varepsilon_i$$

2. Regress the squared residuals on the independent variables:

$$\hat{\nu}_i^2/\hat{\varepsilon}_i^2 = \gamma_0 + \gamma_1 * X_1 + \gamma_2 * X_2 + \dots + \gamma_k * X_k + \nu_i/\varepsilon_i$$

3. LM test:

$$LM = n * R^2$$

or alternatively

$$LM = \frac{(TSS - RSS)}{2}$$

If $LM \leq \chi_{k,\alpha}^2 \Rightarrow$ Homoscedasticity

11.2.7 White Test

involves regressing the squared residuals from the original regression model against the original independent variables and their squares and cross-products. If the coefficients of these regressors are significantly different from zero, it suggests that heteroscedasticity is present.

1. Run the original regression model:

$$Y = \beta_0 + \beta_1 * X_1 + \beta_2 * X_2 + \dots + \beta_k * X_k + \nu_i / \varepsilon_i$$

2. Auxiliary regression of the

$$\nu_i^2 / \varepsilon_i^2 = \gamma_0 + \gamma_1 * X_1 + \gamma_2 * X_2 + \dots + \gamma_k * X_k + \gamma_{k+1} * (X_1 * X_2) + \gamma_{k+2} * (X_1^2) + \gamma_{k+3} * (X_2^2) + \dots + \gamma_m * (X_k^2) + \nu_i / \varepsilon_i$$

3. Perform an F -test or a chi-square test on the coefficients (γ_i) , usually using the residual sum of squares from the auxiliary regression compared to the original model.

If the test statistic is significant, it indicates the presence of heteroscedasticity in the original model.

11.3 Remedies for Heteroscedacity

11.3.1 Generalized Least Squares (GLS)

If the heteroscedastic variances are known, or if the cause of observed heteroscedasticity is known, we can employ Generalized Least Squares (GLS) for more efficient estimates.

Model Specification: Assume that the error variance is related to another variable z_t as follows:

$$\text{Var}(\nu_t / \varepsilon_t) = \sigma^2 z_t^2$$

If we divide the regression equation by z_t , the model can be expressed as:

$$\frac{y_t}{z_t} = \beta_1 + \beta_2 X_{2t} + \beta_3 X_{3t} + \nu_t / \varepsilon_t$$

Where:

$$Y_i^* = \beta_1^* + \beta_2^* X_2^* + \beta_3^* X_3^* + \nu_i^* / \varepsilon_i^*$$

The transformed error term is given by:

$$\nu_t / \varepsilon_t = \frac{\nu_t / \varepsilon_t}{z_t}$$

Thus, we have:

$$\text{Var}(\nu_t / \varepsilon_t) = \sigma^2 z_t^2 \implies \text{Var}(\nu_t / \varepsilon_t) = \text{Var}\left(\frac{\nu_t / \varepsilon_t}{z_t}\right) = \frac{\sigma^2 z_t^2}{z_t^2} = \sigma^2$$

This indicates that for known z_t , the variances of the disturbances from the new regression equation are homoscedastic:

$$\text{Var}(\nu_t / \varepsilon_t) = \sigma^2$$

Implications: - By transforming the model in this way, we can achieve homoscedasticity in the residuals, allowing for valid hypothesis testing and more reliable coefficient estimates. - GLS provides a method to correct for heteroscedasticity when the structure of the variance is known.

11.3.2 Data Transformation

Transforming the dependent variable (e.g., using logarithmic or square root transformations) can stabilize variance across observations. This may reduce the effects of heteroscedasticity by making relationships more linear.

11.3.3 Using White's Corrections

White's correction (also known as robust standard errors) adjusts the standard errors of estimates to account for heteroscedasticity without changing the coefficient estimates themselves. This makes hypothesis tests more reliable despite the presence of heteroscedasticity.

11.3.4 Removing Outliers

Identifying and removing outliers can help alleviate heteroscedasticity as they could disproportionately affect the variance. Care should be taken to ensure that outliers are genuinely not part of the population being studied.

11.3.5 Huber-White Corrections

Huber-White corrections provide robust standard errors that are less sensitive to violations of the homoscedasticity assumption. They can be applied to the ordinary least squares (OLS) estimates to obtain reliable inference even in the presence of heteroscedasticity.

12 Autocorrelation

or serial correlation occurs when error terms in a regression model are correlated across observations in time or space. Symbolically:

$$\text{cov}(\nu_i, \nu_j | X_i, X_j) = E(\nu_i, \nu_j) = 0$$

Types:

Autoregression (AR) - dependency of the lag in depended - Y_{t-1}, Y_{t-p}

Moving Average (MA) - dependency of the lag in residuals - ν_{t-1}, ν_{t-p}

12.1 Sources of Autocorrelation

12.1.1 Inertia

Inertia refers to the tendency of certain processes to maintain their current state until acted upon by external forces. In time series analysis, this can manifest as autocorrelation, where past values influence current values. For instance, in economic data, consumer behavior might follow trends based on previous consumption, leading to correlated residuals. Understanding inertia helps analysts recognize patterns and temporal dependencies in their data, which are essential for accurately modeling relationships.

12.1.2 Non-stationary

Non-stationary data can significantly contribute to autocorrelation. A time series is non-stationary if its statistical properties, such as mean and variance, change over time. When non-stationarity is present, the residuals from a regression model can exhibit correlation, often leading to misleading conclusions. Analysts must test for stationarity using methods such as the Augmented Dickey-Fuller test and apply differencing or detrending techniques to stabilize the series, thereby mitigating the risk of autocorrelation.

12.1.3 Data Transformation

Improper data transformation can introduce autocorrelation into a regression model. If transformations meant to stabilize variance or adjust for non-linearity are misapplied, they may fail to eliminate correlations in the residuals. For example, failing to logarithmically transform exponentially growing data can lead to increased autocorrelation. Careful evaluation of data transformation methods is essential to address underlying correlations effectively and maintain model integrity.

12.1.4 Manipulation of Data

Manipulation of data during the analysis process can lead to autocorrelation, particularly if temporal ordering is disrupted. For example, aggregating data points over time or filtering certain observations may inadvertently create correlation in the residuals. Moreover, if analysts selectively include or exclude data based on outcomes, this can lead to biased results. Adhering to rigorous data management protocols is imperative to avoid introducing autocorrelation through improper data handling.

12.1.5 Specifiaction bias

Specification bias arises when analysts incorrectly model the relationship between independent and dependent variables, often leading to biased estimates and correlated residuals. This can occur if important variables are omitted or inappropriate functional forms are used. For example, failing to include lagged variables in a time series model can result in autocorrelation, as the model does not account for past values affecting future states. Proper model specification is critical to minimizing the risk of autocorrelation and ensuring valid inference.

12.1.6 Cobweb Phenomenon

The cobweb phenomenon describes a situation in dynamic systems where supply and demand adjustments lead to oscillations over time. In economic contexts, this can create autocorrelation in time series data, as each period's supply reaction is based on the prior period's price signals. This cyclical behavior can lead to correlated residuals that violate regression assumptions. Recognizing the cobweb phenomenon allows analysts to anticipate potential autocorrelation patterns and adjust their models accordingly.

12.1.7 Lags

The inclusion of lagged variables can be both a source and a remedy for autocorrelation, depending on how they are implemented in the model. If relevant lagged variables are omitted, the residuals may exhibit autocorrelation, as past values affecting the current outcome are excluded. Conversely, appropriately incorporating lagged variables can help capture temporal dependencies, reducing autocorrelation. Analysts must carefully evaluate the role and significance of lags in their models to ensure the accuracy of results and interpretations.

12.2 Detection of Autocorrelation

12.2.1 A priori

A priori methods for detecting autocorrelation rely on theoretical insights and previous research indicating potential dependence between residuals. Analysts can anticipate autocorrelation based on the characteristics of the data or the processes underlying the model. For example, if the data represents a time-dependent phenomenon, such as economic indicators or temperature readings, one might expect past values to influence current ones. This expectation can guide the selection of appropriate regression techniques and inform the use of specific tests, such as the Durbin-Watson statistic, to formally assess autocorrelation in the residuals.

12.2.2 Graphical method

Graphical methods are effective and intuitive tools for visually detecting autocorrelation in regression models. Common approaches include plotting the residuals against time or constructing an autocorrelation function (ACF) plot. In a time plot, residuals should display a random scatter around zero, indicative of independence. However, systematic patterns or correlations in the plot suggest the presence of autocorrelation. An ACF plot displays the correlation of residuals with their lagged values; significant spikes at certain lags reveal autocorrelation. These visual methods provide immediate insight into the presence and nature of autocorrelation, allowing analysts to take appropriate corrective measures.

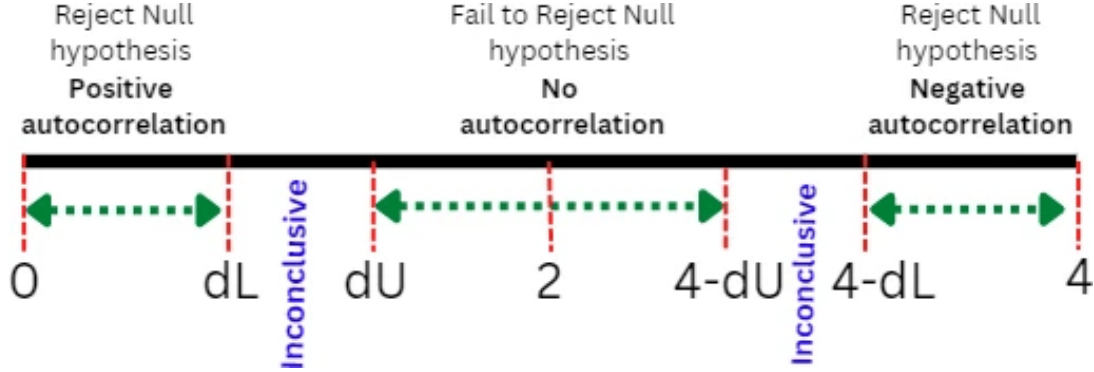
12.2.3 Durbin Watson Test

Assumptions:

- no missing observations
- Intercept is present
- X factors are non-stochastic
- No lag values of Y in the model
- AR(1) only
- Residuals are normally distributed

$$d = \frac{\sum_{t=2}^n (\hat{v}_t - \hat{v}_{t-1})^2}{\sum_{t=1}^n \hat{v}_t^2}$$

Depended on the confidence intervals for reliability of the model for conclusive intervals



Source: reneshbedre.com

12.2.4 Breusch-Godfrey Test

also known as the **LM (Lagrange Multiplier) test**, is a statistical tool used to detect the presence of **autocorrelation** in the residuals of a regression model.

1. Run the original regression model:

$$Y = \beta_0 + \beta_1 * X_1 + \beta_2 * X_2 + \dots + \beta_k * X_k + \nu_i / \varepsilon_i$$

2. Regress the residuals on the lagged residuals variables:

$$\hat{\nu}_i / \hat{\varepsilon}_i = \gamma_0 + \gamma_1 * X_1 + \gamma_2 * X_2 + \dots + \gamma_k * X_k + \varphi_1 * \hat{\nu}_{t-1} / \hat{\varepsilon}_{t-1} + \dots + \varphi_p * \hat{\nu}_{t-p} / \hat{\varepsilon}_{t-p} + \nu_i / \varepsilon_i$$

3. LM test:

$$LM = (n - p) * R^2$$

$$\text{If } LM \leq \chi^2_{p,\alpha} \Rightarrow \text{No autocorrelation.}$$

12.3 Remedies for Autocorrelation

12.3.1 Dynamic Structure

A dynamic model incorporates lagged variables as predictors. This allows the model to account for previous values, potentially reducing autocorrelation by acknowledging the time-dependent structure inherent in the data. For example, including the lagged dependent variable can capture the dynamics better and provide a more accurate representation of the relationship over time.

12.3.2 Generalized Least Squares

effectively transforms the model to account for autocorrelation in the residuals, enabling more efficient and unbiased parameter estimates. By including the lagged dependent variable (Y_{t-1}) the model captures inherent temporal dynamics. The lag of the independent variable X and its respective coefficient allow for the incorporation of changes in X relative to its previous value.

$$\begin{aligned} Y_t - \varphi * Y_{t-1} &= \beta_1 * (1 - \varphi) + \beta_2 * (X_t - \varphi * X_{t-1}) + \nu_t / \varepsilon_t \\ \Rightarrow Y_t &= \varphi * Y_{t-1} + \beta_1 + \beta_2 * (X_t - \varphi * X_{t-1}) + \nu_t / \varepsilon_t \end{aligned}$$

12.3.3 Data Transformation

Common data transformation remedies/methods:

- Logging

- First Differences

- Estimating the residuals

12.3.4 Newey-West Estimates

is a robust method designed to address both autocorrelation (the correlation of error terms across time) and heteroskedasticity (the non-constant variance of error terms) in regression models. It uses quadratic kernel weighting, applying stronger weights to error that are closer in time while diminishing the influence of those that are further away, this helps accurately estimating the long-run covariance of the error terms.

13 Model Specification & Diagnostic Testing

Econometric modeling is an iterative process.

Data $\rightarrow$ Theory $\rightarrow$ Model $\rightarrow$ Diagnostics $\rightarrow$ Respecification

$\circlearrowleft$ Theory $\rightarrow$ Estimation $\rightarrow$ Diagnostics $\rightarrow$ Theory revision $\rightarrow$ Theory $\circlearrowleft$

13.1 Model Selection Criteria

13.1.1 Data Admissible

The model should be built on data that is relevant and permissible for the research question at hand. This criterion emphasizes the importance of using appropriate data types, ensuring that the data is both available and suitable to address the specific analytical objectives. Careful consideration of data sources, quality, and completeness is essential for an effective model.

13.1.2 Concise with Theory

This means that the chosen variables and their relationships should reflect existing theories or empirical findings in the field. A concise model that adheres to theoretical expectations enhances interpretation and supports the validity of the results, aiding in the derivation of meaningful conclusions from the analysis.

13.1.3 Weakly exogenous X factors

meaning that they are determined outside the model and not influenced by the dependent variable. This property is crucial for establishing clear causal interpretations. Weak exogeneity allows for valid inferences about the relationship between the dependent variable and the independent variables, thus ensuring robust parameter estimation.

13.1.4 Parameter Constancy

refers to the assumption that the relationships between variables remain stable over time or across different contexts. For a model to be reliable, its parameters should not vary significantly with changes in the data or the environment. Testing for parameter constancy is essential, as unstable parameters can lead to incorrect predictions and flawed conclusions. Techniques such as Chow tests can help assess this stability.

13.1.5 Data Coherency

denotes the consistency and logical agreement among the data points used in the model. The model should reflect a coherent narrative, avoiding contradictions in the underlying assumptions, data sources, and analytical methods. Ensuring coherence promotes a logical flow in interpretation and enhances the model's credibility.

13.1.6 Encompassing

model is one that incorporates all relevant variables and potential relationships, providing a holistic view of the data. This model should be capable of accommodating various specifications while effectively capturing underlying dynamics. Encompassing ensures that important factors are not left out, and it allows for a comprehensive analysis that can inform better decision-making.

13.2 Specification errors

13.2.1 Underfitting

happens when a model is too simple to capture the underlying relationships in the data. This can occur, for instance, when crucial variables are omitted or when the model is restricted to linear relationships only. As a result, the model fails to explain the variation in the dependent variable adequately, leading to systematic errors in predictions and reduced explanatory power.

13.2.2 Overfitting

occurs when a model becomes excessively complex, capturing not only the underlying relationship but also the noise in the data. This typically arises when too many parameters are included relative to the amount of data available. While an overfitted model may perform exceptionally well on the training data, its predictive power for new or unseen data is often poor. Overfitting can be identified through validation techniques, such as cross-validation, which help to assess the model's performance on unseen data and highlight discrepancies between training and validation metrics.

13.2.3 Wrong Functional Form

Using the incorrect functional form of the relationship between variables can lead to specification errors. For instance, assuming a linear relationship when the true relationship is nonlinear can cause significant bias in the estimates. This misrepresentation may obscure genuine patterns in the data and yield misleading conclusions about the influences of independent variables on the dependent variable. Exploring various functional forms and transformations can help identify a more appropriate model structure.

13.2.4 Errors of Measurement

arise when the dependent or independent variables are measured inaccurately. Such inaccuracies can distort the coefficients and lead to biased or inconsistent estimates. For example, if a survey response inaccurately captures an income variable, the model's ability to infer economic relationships will be compromised. Implementing rigorous data collection methods and validation checks can help mitigate such errors.

13.2.5 Incorrect specification of the error term

refers to failures in modeling the residuals properly. This includes assuming that errors are normally distributed when they are not, neglecting heteroskedasticity, or ignoring autocorrelation in time series data. Failing to address these issues can lead to biased estimates and incorrect statistical inference. Utilizing statistical tests and diagnostic measures can help identify and correct for these errors, ultimately enhancing the reliability of the model's estimates.

13.3 Detection of specification errors

13.3.1 Ramsey's RESET test

(Regression Equation Specification Error Test) is designed to evaluate whether the functional form of a regression model is correctly specified. In particular, it tests for omitted variables or incorrect functional forms. A significant test result may indicate that the model is mis-specified and may need additional variables or a different functional form. How it is computed:

1. Compute original regression model

$$Y_i = \beta_1 + \beta_2 * X_2 + \dots + \beta_k * X_k + \nu_i / \varepsilon_i$$

2. Auxiliary regression with inclusion of fitted Y powers into the regression

$$Y_i = \beta_1 + \beta_2 * X_2 + \dots + \beta_k * X_k + \beta_{k+1} * \hat{Y}^2 + \dots + \beta_{k+m} * \hat{Y}^m + \nu_i / \varepsilon_i$$

3. Adding or removing each X to the original model, one by one in the auxiliary ones.
4. Calculation of the F test

$$F = \frac{\frac{R_{new}^2 - R_{old}^2}{\text{number of new regressors}}}{\frac{1 - R_{new}^2}{n - \text{number of parameters in the new model}}}$$

13.3.2 Lagrange Multiplier (LM) Test for Adding Variables

is a statistical test used to evaluate whether the addition of one or more independent variables improves the fit of a regression model. It is particularly useful in determining if a more complex model is necessary without immediately fitting that model.

1. Compute original regression model

$$Y_i = \beta_1 + \beta_2 * X_2 + \dots + \beta_k * X_k + \nu_i / \varepsilon_i$$

2. Fit the Unrestricted Model

$$Y_i = \beta_1 + \beta_2 * X_2 + \dots + \beta_k * X_k + \beta_{k+1} * X_{k+1} + \dots + \beta_{k+m} * X_{k+m} + \nu_i / \varepsilon_i$$

3. Calculate the LM Statistic:

$$LM = n * (R_{rest}^2 - R_{unrest}^2) \Rightarrow \frac{\hat{\nu}^T * \hat{\nu}}{\sigma^2}$$

13.4 Model Selection Criteria

13.4.1 R^2 Criterion

or coefficient of determination, quantifies how much of the variation in the dependent variable can be explained by the independent variable(s). It is essential in evaluating the goodness-of-fit of regression models.

$$R^2 = 1 - \frac{RSS}{TSS} \Rightarrow \frac{RSS}{ESS}$$

13.4.2 Adjusted R^2

accounts for the number of predictors in the model. Unlike R^2 , which can increase with the addition of irrelevant predictors, adjusted R^2 will only increase if the new predictor improves the model's accuracy. Penalty for overfitting

$$\text{Adj. } R^2 = 1 - \frac{(1 - R^2) * (n - 1)}{n - k - 1} \Rightarrow \frac{\frac{RSS}{n - k - 1}}{\frac{TSS}{n - 1}} \Rightarrow \frac{ESS - k}{n - k - 1}$$

13.4.3 Akaike's Information Criterion (AIC)

is a measure of the relative quality of statistical models for a given dataset. It is used to compare models, with lower values indicating a better fit of the model to the data while penalizing for complexity.

$$AIC = e^{\frac{2k}{n}} * \frac{RSS}{n}$$

Linearized form:

$$\ln AIC = \left(\frac{2k}{n}\right) + \ln\left(\frac{RSS}{n}\right)$$

13.4.4 Schwarz's Information Criterion (SIC)

Also known as the Bayesian Information Criterion (BIC), SIC is similar to AIC but imposes a heavier penalty for models with more parameters. It is particularly useful in model selection when the sample size is large.

$$SIC = n^{\frac{k}{n}} * \frac{RSS}{n}$$

Linearized form:

$$\ln SIC = \left(\frac{k}{n}\right)\ln(n) + \ln\left(\frac{RSS}{n}\right)$$

13.4.5 Mallows's Cp Criterion

provides a measure to assess the fit of a regression model while accounting for the number of predictors. A lower Cp value indicates a model that has a good balance between goodness-of-fit and model simplicity.

$$C_p = \frac{RSS + 2k * \sigma^2}{n}$$

14 Qualitative response regression models

For these reasons John Aldrich and Forrest Nelson contend that “use of the coefficient of determination as a summary statistic should be avoided in models with qualitative dependent variable[s].

14.1 Linear Probability Model

$$Y_i = \beta_1 + \beta_i * X_i + \nu_i / \varepsilon_i$$

because it is conditional probability $\Rightarrow E(Y_i|X_i) = \beta_1 + \beta_i * X_i = P_i$

This model follows the Bernoulli probability distribution:

$$E(Y_i) = 0 * (1 - P_i) + 1 * (P_i) = P_i$$

Limits of the distribution: $0 \leq E(Y_i|X_i) \leq 1$

X_i follows the binomial distribution:

with a Probability Mass Function of $P(X = k) = \binom{n}{k} p^k * (1-p)^{n-k}$

mean of : $\bar{X}_i = n * p$

Variance of : $\sigma^2(X_i) = n * p(1 - p)$

Until the availability of readily accessible computer packages to estimate the logit and pro-bit models (to be discussed shortly), the LPM was used quite extensively because of its simplicity.

14.2 Logit

The logit function is a mathematical transformation primarily utilized in logistic regression, which converts probabilities (ranging from 0 to 1) into log-odds, allowing for effective modeling of binary outcomes. Estimating the parameters of this model typically employs maximum likelihood estimation (MLE), which determines the values that maximize the likelihood of observing the given data. Various optimization methods, such as gradient descent and the Newton-Raphson method, are used to find these estimates efficiently.

$$P_i = \frac{e^{(\beta_1 + \beta_i * X_i)}}{1 + e^{(\beta_1 + \beta_i * X_i)}}$$

The (cumulative) logistic distribution function:

$$P_i = \frac{1}{1 + e^{-(\beta_1 + \beta_i * X_i)}}$$

Reverse probability:

$$\begin{aligned} 1 - P_i &= \frac{1}{1 + e^{(\beta_1 + \beta_i * X_i)}} \\ \Rightarrow \frac{P_i}{1 - P_i} &= \frac{1 + e^{(\beta_1 + \beta_i * X_i)}}{1 + e^{-(\beta_1 + \beta_i * X_i)}} = e^{(\beta_1 + \beta_i * X_i)} \end{aligned}$$

Estimation of the odds ratio:

$$\ln\left(\frac{P_i}{1 - P_i}\right) = \beta_1 + \beta_i * X_i + \nu_i / \varepsilon_i$$

14.3 Probit

known also as normit function is a statistical transformation commonly employed in binary outcome models, offering an alternative to logistic regression. It converts probabilities (ranging from 0 to 1) into z-scores from a standard normal distribution, allowing for the modeling of probabilities based on cumulative density functions. The parameters of this model are typically estimated using maximum likelihood estimation (MLE), similar to logistic regression. Optimization methods, such as gradient descent and the Newton-Raphson method, are utilized to find the estimates efficiently.

Probability function:

$$P_i = P(I_i^* \leq I_i) = P(Z_i \leq \beta_1 + \beta_i * X_i) = F(\beta_1 + \beta_i / X_i), \quad Z_i \sim N(0, \sigma^2)$$

Inverse CDF for Estimation:

$$I_i = F^{-1}(P_i) = \beta_1 + \beta_i * X_i$$

CDF Calculation:

$$F(I_i) = \frac{1}{2} + \frac{\text{Error function}}{2} * \left(\frac{I_i}{\sqrt{2}}\right)$$

14.4 Tobit - Censored Regression model

An extension of the probit model is the tobit model originally developed by James Tobin, the Nobel laureate economist. To explain this model, we continue with our home ownership example. In the probit model our concern was with estimating the probability of owning a house as a function of some socioeconomic variables. In the tobit model our interest is in finding out the amount of money a person or family spends on a house in relation to socioeconomic variables. The Tobit model is utilized in situations where the dependent variable is censored, meaning that it is only observable within certain limits. This model is particularly useful for analyzing outcomes where the values may not vary freely due to constraints.

Latent Variable Equation:

$$Y_i^* = \beta_0 + \beta_i * X_i + \varepsilon_i$$

For censored observations:

$$P(Y_i = 0) = P(Y_i^* \leq 0) = \phi(-(\beta_0 + \beta_i * X_i))$$

For uncensored observations:

$$P(Y_i = Y_i^*) = \phi(Y_i^* | \beta_0 + \beta_i * X_i)$$

14.5 Poisson

There are many phenomena where the regressand is of the count type, such as the number of vacations taken by a family per year, the number of patents received by a firm per year, the number of visits to a dentist or a doctor per year, the number of visits to a grocery store per week, the number of parking or speeding tickets received per year, the number of days stayed in a hospital in a given period, the number of cars passing through a toll booth in a span of, say, 5 minutes, and so on. Just as the Bernoulli distribution was chosen to model the yes/no decision in the linear probability model, the probability distribution that is specifically suited for count data is the Poisson probability distribution.

Probability density function:

$$f(Y_i) = \frac{\mu^{Y_i} * e^{-\mu}}{Y_i!} + \varepsilon_i$$

Both the mean $\bar{Y}$ and the variance σ_2 are the same value of μ .

14.6 Hypothesis testing

using for each coefficient the t-test for small number ($n \leq 30$) of observations & z-test for the bigger number of observation (following the Normal distribution).

14.7 Goodness of fit

14.7.1 McFadden's Pseudo R^2

$$R^2 = 1 - \frac{\text{Log Likelihood (fitted model)}}{\text{Log Likelihood (null model)}}$$

14.7.2 -2 Log Likelihood

Closer to 0 the better fit of the model.

$$-2 * \log(\text{ML Estimate})$$

14.7.3 Hosmer-Lemeshow

The primary goal of the Hosmer-Lemeshow test is to determine if the predictions made by the logistic regression model significantly deviate from the actual outcomes. A good model fit will show that the predicted probabilities are close to the observed frequencies of the binary outcomes (e.g., success vs. failure).

Steps:

1. Data patroning - Dataset is divided into g groups (usually 10 bins) based on predicted probabilities.
2. Calculate Expected & Observed Outcomes -

$$O_i = \text{Success events}$$

$$E_i = \text{expected number of events}$$

3. Formula

$$HL = \sum_{i=1}^g \frac{(O_i - E_i)^2}{E_i * (1 - \hat{p}_i)}$$

Good fit model (Alternative Hypothesis) : $HL \sim \chi^2$ distribution with $(g-2)$ & g degrees of freedom.

14.8 Classification

In the context of models like Logit and Probit, classification metrics help evaluate the model's performance in predicting binary outcomes.

14.8.1 Sensitivity (True Positive Rate)

- **Definition:** The proportion of actual positive cases that are correctly identified by the model.
- **Formula:**

$$\text{Sensitivity} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

14.8.2 Specificity (True Negative Rate)

- **Definition:** The proportion of actual negative cases that are correctly identified by the model.
- **Formula:**

$$\text{Specificity} = \frac{\text{True Negatives (TN)}}{\text{True Negatives (TN)} + \text{False Positives (FP)}}$$

14.8.3 False Positive Rate (FPR)

- **Definition:** The proportion of actual negative cases that are incorrectly classified as positive.
- **Formula:**

$$\text{FPR} = \frac{\text{False Positives (FP)}}{\text{True Negatives (TN)} + \text{False Positives (FP)}}$$

14.8.4 False Negative Rate (FNR)

- **Definition:** The proportion of actual positive cases that are incorrectly classified as negative.
- **Formula:**

$$\text{FNR} = \frac{\text{False Negatives (FN)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

14.8.5 Accuracy

- **Definition:** The proportion of total correct predictions (both positive and negative) out of all predictions made.
- **Formula:**

$$\text{Accuracy} = \frac{\text{True Positives (TP)} + \text{True Negatives (TN)}}{\text{Total Observations}}$$

14.8.6 Cutoff point

In classification models, the **cutoff** (or threshold) value is the probability threshold above which the prediction is considered a positive outcome (e.g., happy) and below which it is considered a negative outcome (e.g., unhappy). Adjusting this threshold can influence the balance of sensitivity and specificity:

- **Lowering the Cutoff:** Increases sensitivity but may decrease specificity, resulting in more true positives but also more false positives.
- **Raising the Cutoff:** Increases specificity but may decrease sensitivity, resulting in fewer false positives but also more false negatives.

Finding the optimal cutoff involves trade-offs based on the specific context, such as prioritizing sensitivity in a medical diagnosis (missing a disease is more critical) or specificity in fraud detection (minimizing false alarms).

14.8.7 Receiver Operating Characteristic (ROC) Curve

The **ROC curve** is a graphical representation that illustrates the diagnostic ability of a binary classifier system as its discrimination threshold is varied.

- **Axes:**
 - The x-axis represents the **False Positive Rate (FPR)**.
 - The y-axis represents the **True Positive Rate (Sensitivity)**.

- **Curve Characteristics:**

- The ROC curve plots the sensitivity against the FPR for various cutoff values.
- A curve that bows towards the upper left corner indicates a better-performing model.

14.8.8 Area Under the Curve (AUC)

AUC is the area under the ROC curve and serves as a summary measure of the model's performance:

- **Interpretation:**

- AUC ranges from 0 to 1:

An **AUC of 0.5** indicates no discriminative ability (performance equivalent to random guessing).

An **AUC of 1.0** indicates perfect discrimination (the model correctly classifies all positive and negative cases).

Higher AUC values suggest better model performance in distinguishing between the two classes.

15 Panel data regression models

Panel data refers to a data set that contains observations on multiple entities (such as individuals, firms, or countries) over multiple time periods. It combines features of both time series data and cross-sectional data, allowing researchers to examine changes within entities and across time. This type of data is valuable for analyzing complex behaviors and relationships since it can capture individual heterogeneity and dynamic changes

15.1 Types of Panel Data

15.1.1 Time Series cross section data

Time series cross-section data involves observations for multiple entities at various time periods. This structure allows for the analysis of trends and patterns both across different entities at a particular time and across time for the same entity. For instance, a researcher might study economic indicators of several countries over a span of years, which enables analysis of individual country trends and comparisons between them.

15.1.2 Micropanel data

Micropanel data contains a relatively small number of observations for a larger number of entities. Often, this type of data focuses on high-frequency measures over a shorter time frame, allowing for nuanced insights into behaviors at the individual level. This can be particularly useful for studies in social sciences, such as examining consumer behavior over a series of shopping trips, providing detailed insights into individual-level change.

15.1.3 Cohort analysis

Cohort analysis focuses on a specific group of entities that share a common characteristic or experience within a defined time frame. By tracking these cohorts over time, researchers can identify patterns and outcomes unique to that group. For example, tracking a cohort of individuals born in the same year enables analysis of their socio-economic mobility and life events across their lifetimes, providing invaluable insights into generational effects.

15.1.4 Event history analysis

Event history analysis (or survival analysis) involves studying the timing and occurrence of specific events among entities over time. This type of analysis is particularly useful for understanding phenomena such as job changes, medical events, or other life transitions. By combining information about individual characteristics and the timing of events, researchers can model the likelihood of certain events occurring and identify factors that influence these occurrences.

15.2 Limitations

15.2.1 Data availability

Often, access to a comprehensive dataset that spans both time and various entities can be challenging. Limited availability of high-quality longitudinal data may hinder the ability to conduct robust analyses. In some cases, researchers may need to rely on incomplete datasets, which can affect the generalizability and reliability of their findings.

15.2.2 Measurement Error

Measurement error can lead to biased estimates in panel data analysis. When the variables of interest are inaccurately captured due to flaws in data collection methods, the resulting inference may distort

true relationships. This issue is particularly problematic in longitudinal studies where the accuracy of measurements may vary over time, complicating the interpretation of results and potentially leading to erroneous conclusions.

15.2.3 Short time dimension

in panel datasets can significantly limit the analytical power. When the time periods are brief, it may not capture enough variability or change within entities, impairing the ability to draw meaningful conclusions. Inadequate time series data can also restrict the examination of long-term trends and relationships, reducing the effectiveness of the study.

15.2.4 Serial Correlation & Heteroskedasticity

In panel data, serial correlation (where residuals are correlated across time) and heteroskedasticity (variation of errors across entities or time) can both bias standard errors and affect the validity of inferential statistics. These issues complicate the modeling process, often necessitating more sophisticated methods to correct for them. Ignoring these aspects may lead to invalid inferences and misleading results.

15.2.5 Model selection uncertainty

is a significant challenge when analyzing panel data. Researchers often have multiple modeling approaches available, and choosing the wrong one can lead to biased or inconsistent estimates. This uncertainty can originate from various sources, such as omitted variable bias, or poor specification of the model structure, making it vital for researchers to rigorously evaluate their model choices and assumptions.

15.2.6 Bias results

can occur in panel data analysis when there are systematic differences between units that enter or leave the panel and those that remain or do not enter at all. This phenomenon often arises in longitudinal studies where the composition of the sample changes over time, leading to issues of sample selection bias.

15.3 Estimation Approaches

15.3.1 Same Intercepts & Slopes

This approach simplifies analysis by treating the sample as a single entity, making it easier to interpret results. However, this can overlook individual heterogeneity, potentially masking unique effects that different entities experience. Consequently, while it may enhance computational efficiency, it risks producing misleading conclusions if significant variations exist among the units.

$$Y_{i,t} = \beta_0 + \beta_1 * X_{1,i,t} + \dots + \varepsilon_{i,t}$$

15.3.2 Variable Intercepts across individuals & Constant Slopes

This estimation approach recognizes the possibility that different entities start from different points (i.e., have different intercepts) based on their unique characteristics. By varying intercepts, researchers can more accurately capture the individual differences, such as background factors or initial levels of the dependent variable. However, assuming constant slopes may still be a limitation if the impact of independent variables varies significantly across individuals.

$$Y_{i,t} = \beta_{0,i} + \beta_1 * X_{1,i,t} + \dots + \varepsilon_{i,t}$$

15.3.3 Variable Intercepts across individuals & time, & Constant Slopes

This approach recognizes that both the baseline levels and the underlying context of each individual can change over time, providing a more flexible framework for analysis. By capturing variations in the intercept, researchers can account for temporal effects, such as economic conditions or policy changes that influence all entities differently at different times. Nevertheless, while this model improves accuracy, it still holds the slope coefficients constant across entities, which may not be appropriate in dynamic contexts.

$$Y_{i,t} = \beta_{0,i,t} + \beta_1 * X_{1,i,t} + \dots + \varepsilon_{i,t}$$

15.3.4 Assume all coefficients vary over individuals

This model captures the full range of individual heterogeneity, acknowledging that different entities may respond differently to changes in independent variables. This approach can be particularly useful when there is reason to believe that the impact of predictors varies significantly among subject groups. However, it can also lead to increased complexity and difficulty in estimation, as the number of parameters increases. The richer model allows for a more nuanced understanding of the data, but care must be taken to avoid overfitting.

$$Y_{i,t} = \beta_{0,i,t} + \beta_{1,i} * X_{1,i,t} + \dots + \varepsilon_{i,t}$$

15.3.5 Assume all coefficients vary over individuals & time

In this model, both the intercepts and the slopes can change across different entities and time periods, allowing for a detailed and sophisticated analysis of dynamic behaviors. This extensive flexibility enables researchers to explore complex relationships and account for varying responses to independent variables across the panel. However, the trade-off is that this approach requires substantial data and computational resources to estimate, and there is a risk of model overfitting if the complexity is not justified by the data.

$$Y_{i,t} = \beta_{0,i,t} + \beta_{1,i,t} * X_{1,i,t} + \dots + \varepsilon_{i,t}$$

15.4 Fixed Effect Model

The Fixed Effect Model (FEM) is a popular estimation technique used in panel data analysis to control for unobserved heterogeneity when this unobserved effect is constant over time. This method specifically accounts for omitted variable bias by focusing on variations within the same unit across time, rather than comparing different units. The fundamental assumption is that individual-specific effects are correlated with the independent variables, and thus, by analyzing each unit individually, we can eliminate these biases.

$$Y_{i,t} = \beta_0 + \beta_1 * X_{1,i,t} + \dots + \alpha_i + \varepsilon_{i,t}$$

Here, α_i represents the individual fixed effect for entity i , capturing all unobserved, time-invariant characteristics specific to that entity. This allows the analysis to focus solely on the relationship between the independent variables X and the dependent variable Y without the influence of these unobserved factors.

15.5 Random Effect Model

The Random Effect Model (REM) is another key estimation technique used in panel data analysis. Unlike the Fixed Effect Model, which assumes that individual-specific effects are correlated with the independent variables, the Random Effect Model assumes that these effects are uncorrelated. This

fundamental difference allows REM to include both within and between variations in the data, facilitating the estimation of effects for time-invariant variables.

$$Y_{i,t} = \beta_0 + \beta_1 * X_{1,i,t} + \dots + \mu_i + \varepsilon_{i,t}$$

In this formulation, μ_i represents the random effect for entity i , which captures the unobserved individual-specific characteristics that are assumed to be randomly distributed across entities. This allows the model to estimate coefficients for both time-varying and time-invariant variables simultaneously.

15.6 FEM vs REM

Feature	Fixed Effect Model (FEM)	Random Effect Model (REM)
Assumption of Effects	Assumes individual-specific effects are correlated with the independent variables.	Assumes individual-specific effects are uncorrelated with the independent variables.
Inclusion of Variables	Cannot estimate coefficients of time-invariant variables.	Can include both time-varying and time-invariant variables in the analysis.
Focus	Analyzes within-unit variations over time.	Analyzes both within and between variations across units.
Estimation Efficiency	Less efficient, especially with large datasets, due to loss of degrees of freedom.	Generally more efficient, leading to potentially more precise estimates.
Bias	Controls for unobserved individual-specific factors, reducing potential bias.	Susceptible to bias if the assumption of independence is violated.

Table 4: Comparison of Fixed Effect Model and Random Effect Model

15.6.1 Strengths of FEM

The Fixed Effect Model is particularly beneficial when researchers want to focus on changes within the same unit over time. This model effectively removes the impact of time-invariant characteristics, providing clearer insights into the relationships between variables. For scenarios where omitted variable bias is a significant concern—especially when unobserved effects correlate with the independent variables—FEM is a preferred choice.

15.6.2 Strengths of REM

In contrast, the Random Effect Model is advantageous when researchers believe that individual-specific effects are random and not correlated with the independent variables. This model's ability to include time-invariant variables enhances its versatility. It is particularly useful when there is a need to analyze variations across both entities and time, making it a strong option for broader analyses.

16 Dynamic Econometric Models: Autoregressive & Distributed-Lag Models

Dynamic econometric models are essential for understanding economic behavior over time, particularly because economic actions often reflect inherent delays. These delays manifest in various ways, such as consumption smoothing, investment adjustments, and wage rigidity. For instance, consumers tend to rely on past income trends, affecting their current consumption patterns. Similarly, firms may delay investment decisions to avoid destabilizing their operations, indicating that the economic response is rarely instantaneous.

Furthermore, the implementation and transmission of policies—whether fiscal, monetary, or related to labor markets—are affected by notable delays. Policymakers must understand that the effects of their actions may not materialize immediately, complicating their evaluation of policy efficacy.

Many economic relationships are also shaped by past values of certain variables, which may include habits, expectations, contracts, and adjustment costs. Dynamic models are particularly valuable as they facilitate the distinction between short-run and long-run effects, providing insights into how economic variables adjust over time.

Autoregressive and distributed-lag models serve as structured frameworks to capture inertia, persistence, and the dissemination effects present in economic time series. These models enhance our capacity for proper inference, forecasting, and policy evaluation while allowing for more robust structural interpretations of economic phenomena.

16.1 Distributed Lag Models

Distributed Lag Models are designed to analyze the relationship between a dependent variable and its independent variable(s) across multiple time periods. This model is particularly valuable in fields like economics where the effects of policy changes or external shocks may take time to manifest. By examining how these lagged values contribute to current outcomes, decision-makers can better understand the timing and strength of these effects and make informed predictions regarding future trends.

$$Y_t = \alpha + \beta_0 * X_t + \beta_1 * X_{t-1} + \beta_2 * X_{t-2} + \varepsilon_t$$

16.2 Autoregressive Models

Autoregressive Models focus on predicting the current value of a dependent variable based on its past values along with contemporary influences from other independent variables. This feedback loop allows the model to incorporate the inherent temporal dependence often found in time series data. Autoregressive Models are especially useful in situations where past behaviors significantly influence present outcomes, such as in finance or meteorology. Through this structure, researchers can uncover patterns and trends, enabling more accurate forecasts based on historical behavior, which is essential for strategic planning and analysis.

$$Y_t = \alpha + \beta * X_t + \gamma * Y_{t-1} + \varepsilon_t$$

16.3 Conceptual versus Statistical Lags

16.3.1 Conceptual Lags

Conceptual lags refer to the theoretical delays inherent in economic processes. These delays are rooted in economic behavior and depend on various factors, such as decision-making time frames and market adaptations. For instance, a price change may lead to a delayed consumer response not immediately captured in statistical models, especially if consumers take time to assess changes in the market or their personal circumstances. Conceptual lags can be long and nuanced, reflecting the complexities of human behavior and economic interactions.

16.3.2 Statistical Lags

Statistical lags, on the other hand, are the mathematical representations of these delays. They can be modeled using various structures, such as geometric lag, polynomial lag, or autoregressive (AR) models. While statistical lags aim to capture the effects of past values on present outcomes, they do not necessarily align with the conceptual understanding of economic delays. As such, statistical lags can sometimes oversimplify or misrepresent the true dynamics of economic behavior.

Reasons for the importance of differentiation of the lags:

1. **Overfitting:** Confusing these lags can lead to models that include too many lagged variables, sacrificing generalizability for supposedly greater explanatory power, which complicates the model without adding real value.
2. **Misinterpretation of Effects:** Misunderstanding the relationship between short-run and long-run effects can lead to inaccurate conclusions about the nature of economic relationships. For instance, one might misinterpret transient effects as permanent without recognizing the underlying dynamics.
3. **Incorrect Economic Conclusions:** If econometric models do not align with economic theory, the resulting analysis may lead to flawed policy recommendations or business decisions, undermining the practical applications of the findings.

16.4 Geometric Lag

The Geometric Lag model is a specific type of distributed lag model that assumes the impact of a change in an independent variable diminishes geometrically over time. In this framework, the lagged effects of a variable decrease at a constant rate, leading to a sequence where each successive impact is a constant proportion of the previous one.

$$Y_t = \alpha + \beta_0 * X_t + \beta_1 * X_{t-1} + \beta_2 * X_{t-2} + \dots + \beta_k * X_{t-k}$$

Where:

$$\beta_1 = \beta_0 * \lambda^1$$

$$\beta_2 = \beta_0 * \lambda^2$$

$$\beta_k = \beta_0 * \lambda^k$$

The weights value is $0 < \lambda < 1$, constant decaying effect from

16.5 Koyck transformation

The Koyck Transformation is a method used to model distributed lags in a way that simplifies the estimation of geometric lag relationships. By transforming the original model, this approach allows researchers to capture lagged effects more effectively while making the econometric analysis more tractable.

$$Y_t = \alpha * (1 - \lambda) + \beta_0 * X_t + \lambda * Y_{t-1} + \varepsilon_t$$

$$\varepsilon_t = \varepsilon_t - \varepsilon_{t-1}$$

Median lag for 50% change in Y following a unit change in X

$$\text{Median lag} = -\frac{\ln(2)}{\ln(\lambda)}$$

Mean Time needed for effects on Y to occur:

$$\text{Mean time} = -\frac{\ln(2)}{\ln(\lambda)}$$

16.6 Almon Model

The Almon Approach is a method used in econometrics to impose a polynomial structure on lag coefficients. This allows researchers to reduce the number of parameters in lagged models while capturing economically meaningful shapes of lagged effects.

$$Y_t = \alpha + \sum_{k=0}^K (\beta_k * X_{t-k}) + \varepsilon$$

$$\beta_k = \theta_0 + \theta_1 * k + \theta_2 * k^2 + \dots + \theta_p * k^p$$

In summary, the Almon approach enhances the modeling of lagged effects by allowing for a polynomial structure that captures important economic relationships while simplifying the estimation process. It proves especially useful for scenarios where traditional lag structures may fall short in reflecting complex lagged responses, such as hump-shaped responses (Investments) & delayed reactions (policy).

16.7 Granger Causality

The core idea behind Granger causality is that if a variable X Granger-causes another variable Y , then the past values of X should contain relevant information that helps predict Y . However, Y does not necessarily cause X ; it indicates a predictive relationship.

To assess whether X Granger-causes Y , we typically compare two models:

1. **Restricted Model** (Only past values of Y):

$$Y_t = \alpha + \sum_{i=1}^p \beta_i Y_{t-i} + \varepsilon_t$$

2. **Unrestricted Model** (Past values of both Y and X):

$$Y_t = \alpha + \sum_{i=1}^p \beta_i Y_{t-i} + \sum_{j=1}^q \gamma_j X_{t-j} + \varepsilon_t$$

The test:

1. **Choose Lag Length:** Determine the appropriate number of lags p and q using criteria like the Akaike Information Criterion (AIC) or Bayesian Information Criterion (BIC).
2. **Estimate Both Models:** Estimate the restricted and unrestricted models.
3. **Perform an F-Test:** Compare the restricted and unrestricted models using an F-test:

$$F = \frac{(RSS_R - RSS_{UR})/(k)}{RSS_{UR}/(n - k - m)}$$

Where:

- RSS_R : Residual sum of squares for the restricted model.
 - RSS_{UR} : Residual sum of squares for the unrestricted model.
 - k : Number of restrictions (number of lagged X terms).
 - n : Total number of observations.
 - m : Number of parameters estimated in the unrestricted model.
4. **Decision Rule:** Compare the computed F-statistic to the critical value from the F-distribution. If the F-statistic exceeds the critical value, reject the null hypothesis, indicating X Granger-causes Y .

17 Simultaneous - Equation Models

Simultaneous-equation models (SEMs) describe systems in which multiple endogenous variables interact with each other and are jointly determined. They are common in economics, engineering, and other fields where feedback loops exist.

17.1 Casual & Non Casual Models

17.1.1 Casual Models

Causal models seek to establish cause-and-effect relationships among variables. They are built on the premise that changes in one variable will directly lead to changes in another. This approach is rooted in theories that explain how variables influence one another, often emphasizing the mechanisms that drive these relationships. For example, a causal model might investigate how an increase in education levels can lead to higher income, detailing various pathways and interactions in the process.

Key characteristics of causal models include their structural nature, where equations explicitly specify causal mechanisms. These models incorporate both endogenous and exogenous variables and require a careful consideration of the relationships between them. Structural coefficients derived from these models have economic interpretations, such as elasticities or marginal effects, which provide insights into the strength and nature of the relationships.

However, estimating causal models can be complex, particularly when simultaneity exists (i.e., variables influence each other simultaneously). Such complexity necessitates sophisticated identification strategies, including methods like instrumental variable estimation (IVS), two-stage least squares (TSLS), or full information maximum likelihood (FIML). Despite these challenges, causal models are invaluable for policy simulations and counterfactual analysis, enabling researchers to assess the effects of potential interventions or policy changes.

17.1.2 Non Casual Models

In contrast, non-causal models primarily focus on associations or correlations between variables rather than establishing direct causal relationships. These models are useful for exploring patterns within data and highlighting relationships without inferring causation. Non-causal models often employ regression techniques to examine how various independent variables correlate with a dependent variable without implying a directional influence.

One key class of non-causal models is the reduced form model. While these models may be derived from underlying theoretical frameworks, they do not specify the causal mechanisms. Instead, they present endogenous variables as functions of only exogenous variables, permitting estimation through ordinary least squares (OLS) since the independent variables are not influenced by the dependent variable. This feature makes non-causal models easier to estimate but limits their utility in causal inference.

Non-causal models serve important roles in forecasting and descriptive analytics. They can identify relationships and trends within the data, allowing researchers to understand correlations that may inform further exploration or hypothesis generation. However, because they do not provide insights into causal pathways, their interpretations must be handled with caution, particularly in decision-making contexts where causation is implied.

17.2 Structural Model

are grounded in theoretical frameworks that aim to describe the relationships between variables based on causal mechanisms. They explicitly detail how independent variables influence dependent variables, outlining the pathways of influence. For instance, a structural model might analyze how policy changes affect economic indicators by incorporating various factors that represent consumer behavior or market dynamics. These models emphasize causal relationships, with each equation specifying how one or more independent variables directly affect one or more dependent variables. They also include both endogenous

variables, which are influenced by others in the model, and exogenous variables, which are not. This comprehensive view of the system allows for a deeper understanding of the interactions at play.

However, estimating structural models can be challenging due to simultaneity, where multiple variables may influence one another. Ordinary least squares (OLS) cannot typically be employed in these cases, necessitating the use of advanced identification techniques such as instrumental variables, two-stage least squares, or full information maximum likelihood. Despite these complexities, structural models are particularly valuable for policy simulations and counterfactual analyses, enabling researchers to predict the effects of changes in one variable on others, thus informing decision-making and intervention design.

17.2.1 Non Structural Model

In contrast, non-structural models, often called reduced form models, primarily focus on illustrating the correlations between variables without explicitly describing the causal relationships. These models are less focused on underlying theoretical assumptions and more on observational data. They highlight how dependent variables relate to independent variables through statistical associations, rather than asserting that one directly influences the other. Non-structural models typically only incorporate exogenous variables on the right-hand side, allowing for simpler estimation procedures. This makes them easier to use in practice, suitable for exploratory analysis, and effective in assessing relationships within the data.

However, while non-structural models can reveal patterns and correlations, they do not support causal inference. Consequently, interpretations derived from these models must be approached cautiously, especially in policy contexts where understanding cause-and-effect relationships is crucial.

17.3 Multiple regression versus Structural Model

Multiple regression is a statistical technique used to assess the relationship between one dependent variable and multiple independent variables. It primarily aims to understand the impact of various predictors on the outcome variable while controlling for other factors. This approach often assumes that the relationships among variables are linear, and it is widely implemented for predictive analysis and correlation assessment. The coefficients obtained from a multiple regression analysis indicate the strength and direction of the associations, allowing for interpretable results regarding how changes in independent variables might affect the dependent variable. However, it is crucial to note that multiple regression focuses on associations and does not establish causal relationships, making it necessary to interpret results with caution.

In contrast, structural models are firmly grounded in theoretical frameworks that aim to depict the causal mechanisms underlying relationships among variables. These models explicitly construct equations that illustrate how independent variables affect dependent variables, taking into account the potential interactions and feedback loops that may exist within a system. Unlike multiple regression, structural models incorporate both endogenous and exogenous variables, providing a more holistic view of the relationships at play. The emphasis on causal mechanisms is fundamental; therefore, structural models often require advanced estimation techniques due to challenges such as simultaneity. This complexity allows them to be particularly valuable for policy simulations and counterfactual analyses, as they can estimate the potential impact of hypothetical changes in policy or other interventions based on a deeper understanding of the system dynamics.

While both methodologies are used for analyzing relationships between variables, the main distinction lies in their assumptions and objectives. Multiple regression is suited for exploratory data analysis and predictive modeling but does not address causality. In contrast, structural models focus on unpacking the intricate cause-and-effect relationships and are thus better suited for simulations and understanding the dynamics within a system. The choice between multiple regression and structural models depends largely on the specific research questions and objectives that a study aims to address.

17.4 Structuring the Model

Theory trimming involves refining a statistical model by systematically adding or deleting paths based on their significance and overall model fit. Here's a breakdown of the two primary approaches: top-down and bottom-up.

17.4.1 Top-Down Approach

1. Initial Step: Full Model

- Begin by running the **full model** which includes all potential paths between variables.
- Identify **non-significant paths** that do not contribute meaningfully to the model.

2. Path Removal Process

- **Re-run the model:** Drop one non-significant path at a time and re-evaluate the model.
- **Assess Fit Changes:** Monitor changes in model fit indices (e.g., R^2 , AIC, BIC) after each path removal to determine the contribution of that path.

3. Significance Analysis

- Continuously check the significance of the remaining paths to ensure they provide meaningful insights.
- Focus on the overall improvement in model fit metrics.

17.4.2 Bottom-Up Approach

1. Initial Step: Reduced Model

- Start with a **theoretically grounded reduced model** that includes only the hypothesized significant paths.

2. Adding Paths

- Sequentially **re-run the model** by adding missing paths that were excluded initially.
- Assess whether the addition of these paths improves the model fit.

3. Evaluation Metrics

- Analyze the significance of the newly added paths.
- Evaluate overall model fit improvements using fit indices.

17.4.3 Assessment Criteria

When performing theory-driven modeling, it's essential to look at:

- **Significance of Paths:** Ensure each path in the model has a statistically significant impact.
- **Overall Improvement in Fit:** Measure model fit improvements using metrics such as R^2 , AIC, and BIC to validate the choices made during path addition or deletion.

17.5 Structural Model - issues

17.5.1 Endogenous variables

Endogenous variables are those whose values are influenced by other variables within the model. They are typically dependent variables that can be affected by changes in independent variables. In a structural equation model, an endogenous variable might cause or be caused by other variables, leading to potential issues like multicollinearity or omitted variable bias if not properly addressed.

17.5.2 Exogenous variables

Exogenous variables are independent variables whose values are not affected by other variables in the model. They serve as predictors and are assumed to be unaffected by changes in the model's endogenous variables. Identifying clear exogenous variables is crucial for establishing causal relationships in the model.

17.5.3 Identification

Identification refers to the ability to determine unique parameter estimates within a model. A model is said to be identified if there is sufficient data to estimate all parameters without ambiguity. Poor identification can arise if there are too few observations relative to parameters, leading to non-unique solutions and difficulty in inference.

17.5.4 Estimation

Estimation is the process of calculating the parameters of the model based on data. Common estimation methods include Ordinary Least Squares (OLS), Maximum Likelihood (ML), and Bayesian estimation. The choice of estimation method can significantly impact the robustness and validity of the model results, especially when dealing with endogenous variables or models that are not fully identified.

18 Identification problem

18.1 Endogeneity & Exogeneity

Endogenous variables are determined within the structural system (e.g., Y variables), while exogenous (predetermined) variables are determined outside the system (e.g., X variables) and are not affected by the endogenous variables. Properly distinguishing the two is essential for causal inference.

18.2 Identification Methods

Several approaches are used to achieve identification:

- **Order condition:** For the i -th equation, the number of excluded exogenous variables g_i must satisfy $g_i \geq k_i$, where k_i is the number of endogenous regressors in that equation.
- **Rank condition:** The matrix of coefficients on the excluded exogenous variables must have full rank k_i .
- **Instrumental variables (IV):** Valid instruments are exogenous variables correlated with the endogenous regressors but uncorrelated with the error term.
- **Two-Stage Least Squares (2SLS):** Implements IV estimation for exactly identified or over-identified systems.
- **Generalized Method of Moments (GMM):** Extends IV/2SLS to allow for heteroskedasticity and additional moment conditions.

18.3 Rules for Identification

Practical steps to verify identification in a linear structural model:

1. **Classify variables** as endogenous or exogenous.
2. **Write each structural equation** and list the exogenous variables that are omitted from it.
3. **Apply the order condition:** Check that $g_i \geq k_i$ for every equation.
4. **Construct the instrument matrix $\mathbf{Z}$** using all exogenous variables (both included and excluded) and verify the rank condition:
$$\text{rank}(\mathbf{Z}'\mathbf{X}_i) = k_i$$
where $\mathbf{X}_i$ contains the endogenous regressors in equation i .
5. **Choose an estimation method:**
 - Exactly identified $\rightarrow$ 2SLS gives a unique solution.
 - Over-identified $\rightarrow$ 2SLS or GMM plus an over-identification test (Sargan/Hansen) to assess instrument validity.
6. **Test for weak instruments:** Compute the first-stage F -statistic; values below 10 may indicate weak identification.

Notation

- M – number of endogenous variables in the whole model.
- m – number of endogenous variables that appear in a given equation (excluding the dependent variable).
- K – number of predetermined (exogenous) variables in the model, including the intercept.
- k – number of predetermined variables that appear in that given equation.

Identification condition for simultaneous-equation models In a system of M simultaneous equations, an individual equation is identified only if it excludes at least $M - 1$ variables (both endogenous and predetermined) that appear elsewhere in the system.

If the equation excludes exactly $M - 1$ variables $\Rightarrow$ just identified,

If it excludes more than $M - 1$ variables $\Rightarrow$ overidentified.

Equivalently, using the notation above, the necessary order condition is

$$K - k \geq m - 1$$

where

- $K - k$ = number of predetermined variables excluded from the equation,
- $m - 1$ = number of endogenous variables included in the equation minus one.

If $K - k = m - 1$, the equation is just identified; if $K - k > m - 1$, it is overidentified.

19 Simultaneous - Equation Methods

19.1 Recursive Models

A recursive, or triangular, system can be written as a set of structural equations in which each equation contains only those endogenous variables that have been defined in earlier equations.

$$\begin{aligned}Y_{1t} &= \alpha_1 + \beta'_1 X_t + u_{1t}, \\Y_{2t} &= \alpha_2 + \gamma_{21} Y_{1t} + \beta'_2 X_t + u_{2t}, \\Y_{3t} &= \alpha_3 + \gamma_{31} Y_{1t} + \gamma_{32} Y_{2t} + \beta'_3 X_t + u_{3t}.\end{aligned}$$

Key features

1. ****Ordering**** – Y_1 appears only in the first equation; Y_2 may depend on Y_1 but not on Y_3 ; Y_3 may depend on Y_1 and Y_2 but never on a variable that appears later in the ordering.
2. ****No feedback loops**** – because the coefficient matrices for the endogenous variables are strictly lower-triangular (the γ 's appear only below the main diagonal).
3. ****Reduced form**** – each endogenous variable can be expressed solely in terms of the exogenous variables by successive substitution:

$$\begin{aligned}Y_{1t} &= \alpha_1 + \beta'_1 X_t + u_{1t}, \\Y_{2t} &= \alpha_2 + \gamma_{21}(\alpha_1 + \beta'_1 X_t + u_{1t}) + \beta'_2 X_t + u_{2t}, \\Y_{3t} &= \alpha_3 + \gamma_{31}(\alpha_1 + \beta'_1 X_t + u_{1t}) + \gamma_{32}(\alpha_2 + \gamma_{21} Y_{1t} + \beta'_2 X_t + u_{2t}) + \beta'_3 X_t + u_{3t}.\end{aligned}$$

4. ****Estimation**** – because each equation contains only exogenous regressors (or already-estimated endogenous variables), ordinary least squares applied to each equation yields consistent and unbiased estimates of the α 's, β 's and γ 's.

Thus, a recursive model is a special case of a simultaneous-equation system that can be estimated by OLS without the need for instrumental-variable techniques.

19.2 Indirect Least Squares (ILS)

Indirect Least Squares (ILS) uses the reduced-form equations to back-out the structural parameters. Consider a simple two-equation simultaneous system with endogenous variables Y_1 (demand) and Y_2 (supply) and a set of exogenous (predetermined) variables X .

$$\begin{aligned}(1) \text{ Demand: } & Y_{1t} = \alpha_1 + \beta_{12} Y_{2t} + \gamma'_1 X_t + u_{1t}, \\(2) \text{ Supply: } & Y_{2t} = \alpha_2 + \beta_{21} Y_{1t} + \gamma'_2 X_t + u_{2t}.\end{aligned}$$

Step 1 – Estimate reduced-form equations Write each endogenous variable as a function of only the exogenous variables:

$$\begin{aligned}Y_{1t} &= \delta_{10} + \delta'_{11} X_t + e_{1t}, \\Y_{2t} &= \delta_{20} + \delta'_{21} X_t + e_{2t}.\end{aligned}$$

Run ordinary least squares on the two reduced-form regressions and obtain the estimated coefficients $\hat{\delta}_{10}, \hat{\delta}_{11}, \hat{\delta}_{20}, \hat{\delta}_{21}$.

Step 2 – Express structural coefficients in terms of reduced-form coefficients For the two-equation case the algebraic relationships are

$$\beta_{12} = \frac{\delta_{21}}{\delta_{20}}, \quad \beta_{21} = \frac{\delta_{11}}{\delta_{10}}.$$

More generally, each structural coefficient is a rational function of the reduced-form coefficients; the exact form follows from solving the simultaneous system for the endogenous variables.

Step 3 – Substitute the estimated reduced-form coefficients Replace the population reduced-form coefficients by their OLS estimates:

$$\hat{\beta}_{12} = \frac{\hat{\delta}_{21}}{\hat{\delta}_{20}}, \quad \hat{\beta}_{21} = \frac{\hat{\delta}_{11}}{\hat{\delta}_{10}}.$$

These $\hat{\beta}$'s are the ****Indirect Least Squares**** estimates of the structural parameters.

Remarks

- ILS works directly only when the equation is ****exactly identified****; otherwise the mapping from reduced-form to structural parameters is not unique and additional restrictions are required.
- Because the reduced-form coefficients are obtained by OLS, they are consistent; the subsequent algebraic transformation preserves consistency under exact identification.

19.3 Two-Stage Least Squares (2SLS)

Two-Stage Least Squares yields consistent estimates for every equation of a simultaneous-equation system, whether the system is recursive or not.

1. First stage

For each endogenous regressor X_e regress it on the full set of predetermined (exogenous) variables Z (the instruments). Using OLS obtain the fitted values $\hat{X}_e$:

$$X_e = Z\Pi_e + v_e \implies \hat{X}_e = Z\hat{\Pi}_e.$$

2. Second stage

Substitute the fitted values $\hat{X}_e$ for the endogenous regressors in the structural equation and estimate by OLS:

$$y = X_e\beta + X_c\gamma + u \implies y = \hat{X}_e\beta + X_c\gamma + u,$$

where X_c denotes the exogenous regressors that appear directly in the equation.

Identification remarks

- If the equation is **exactly identified**, 2SLS coincides with the standard instrumental-variables estimator.
- If the equation is **overidentified**, all available instruments are used; a Sargan/Hansen test can be performed to check instrument validity.

20 Time Series Econometrics: Basic Concepts

20.1 Stationary Stochastic Process

A stochastic process $\{Y_t\}$ is (weakly) stationary if its first two moments are time-invariant:

$$\begin{aligned}\mathbb{E}[Y_t] &= \mu \quad (\text{constant mean}), \\ \text{Var}(Y_t) &= \sigma^2 \quad (\text{constant variance}), \\ \text{Cov}(Y_t, Y_{t-h}) &= \gamma(h) \quad \text{depends only on lag } h.\end{aligned}$$

Typical examples are ARMA processes with roots outside the unit circle.

20.2 Non-stationary Stochastic Process

A process is non-stationary when at least one of the above conditions fails. Common forms:

- Time-varying mean (e.g., deterministic trend).
- Time-varying variance (heteroskedasticity).
- Autocovariances that depend on the specific time periods, not just on the lag.

Non-stationarity often leads to unreliable inference if ignored.

20.3 Autocorrelation Function (ACF)

The autocorrelation function measures the linear dependence between observations of a time series at different lags. For a covariance-stationary series $\{Y_t\}$,

$$\rho(k) = \frac{\text{Cov}(Y_t, Y_{t-k})}{\text{Var}(Y_t)} = \frac{\gamma(k)}{\gamma(0)}, \quad k = 0, 1, 2, \dots$$

where $\gamma(k)$ is the autocovariance at lag k . The sample ACF is

$$\hat{\rho}(k) = \frac{\sum_{t=k+1}^T (Y_t - \bar{Y})(Y_{t-k} - \bar{Y})}{\sum_{t=1}^T (Y_t - \bar{Y})^2}.$$

Plotting $\hat{\rho}(k)$ against k helps identify the order of moving-average (MA) components.

20.4 Partial Autocorrelation Function (PACF)

The partial autocorrelation at lag k is the correlation between Y_t and Y_{t-k} after removing the linear effects of the intervening lags $1, \dots, k-1$. It can be obtained from the recursive Yule-Walker equations or by fitting an $\text{AR}(k)$ model and extracting the last coefficient:

$$\phi_{kk} = \text{PACF}(k).$$

For an $\text{AR}(p)$ process, the PACF cuts off after lag p ; for an $\text{MA}(q)$ process, the PACF tails off gradually.

20.5 Ljung–Box Q-Statistic

The Ljung–Box test assesses whether a group of autocorrelations up to lag h are jointly zero. The test statistic is

$$Q(h) = T(T+2) \sum_{k=1}^h \frac{\hat{\rho}^2(k)}{T-k},$$

where T is the sample size and $\hat{\rho}(k)$ are the sample autocorrelations. Under the null hypothesis of no autocorrelation, $Q(h)$ follows an asymptotic χ^2 distribution with $h-p$ degrees of freedom (p = number of parameters estimated in the fitted model). A large $Q(h)$ (small p-value) indicates remaining serial correlation in the residuals.

These tools are essential for diagnosing model adequacy in ARMA modelling and for checking the white-noise assumption of residuals.

20.6 Unit-Root Stochastic Process

A unit-root process is a particular non-stationary process that can be written as

$$Y_t = \rho Y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \text{i.i.d.}(0, \sigma^2),$$

with $\rho = 1$. The series follows a random walk (possibly with drift). Differencing once, $\Delta Y_t = Y_t - Y_{t-1} = \varepsilon_t$, produces a stationary series. Unit-root tests (ADF, PP, KPSS) are used to detect $\rho = 1$.

20.7 Trend (TS) & Difference (DS) Stationary

- **Trend-Stationary (TS):** The series becomes stationary after removing a deterministic trend:

$$Y_t = \beta_0 + \beta_1 t + u_t, \quad u_t \text{ stationary.}$$

Detrending (regressing on time) yields a stationary residual.

- **Difference-Stationary (DS):** The series becomes stationary after differencing (usually first difference):

$$\Delta Y_t = Y_t - Y_{t-1} = u_t, \quad u_t \text{ stationary.}$$

DS series contain a stochastic trend (unit root); differencing removes the stochastic component.

Correctly distinguishing TS from DS is crucial for forecasting and cointegration analysis.

20.8 Spurious Regressions

When two non-stationary series (especially unit-root series) are regressed on each other, the OLS fit often shows a high R^2 and significant t-statistics even though the variables are unrelated. This occurs because the stochastic trends induce a deterministic correlation in levels. Remedies:

- Test each series for unit roots and transform to stationarity (detrend or difference) before regression.
- If both series are integrated of order one, test for cointegration; a valid regression can be run on the cointegrating relationship.

20.9 Tests of Stationarity

20.9.1 Graphical Analysis

Plot the time series, its sample autocorrelation function (ACF), and its partial autocorrelation function (PACF).

- A roughly constant mean and variance, and ACF that dies out quickly, suggest stationarity.
- A clear upward or downward trend, or an ACF that decays very slowly, indicates non-stationarity.

Visual inspection is a useful first step before formal tests.

20.9.2 Dickey–Fuller (DF) Test

The DF test examines the null hypothesis of a unit root in an autoregressive model:

$$\Delta Y_t = \alpha + \beta t + \gamma Y_{t-1} + \sum_{i=1}^p \delta_i \Delta Y_{t-i} + \varepsilon_t.$$

- $\gamma = 0 \Rightarrow$ unit root (non-stationary).
- $\gamma < 0 \Rightarrow$ stationary.

The test statistic is the t -ratio for $\hat{\gamma}$; critical values are non-standard (Dickey–Fuller tables).

20.9.3 Autocorrelation Function (ACF) Test

For a stationary series the sample ACF should lie within the approximate 95% confidence bounds $\pm 1.96/\sqrt{T}$.

- Persistent significant autocorrelations at many lags suggest a unit-root process.
- Rapid decay of autocorrelations supports stationarity.

The Ljung–Box Q -statistic (see previous section) can be applied to a set of lags to test the joint hypothesis that all autocorrelations are zero.

20.9.4 Phillips–Perron (PP) Test

The PP test is a non-parametric modification of the DF test that corrects for serial correlation and heteroskedasticity in the error term without adding lagged difference terms:

$$\Delta Y_t = \alpha + \beta t + \gamma Y_{t-1} + \varepsilon_t.$$

The test statistic is the same t -ratio for $\hat{\gamma}$, but the variance estimator is adjusted using Newey–West type corrections. Critical values are identical to those of the DF test, making PP a convenient alternative when the error structure is unknown.

20.10 Transforming Nonstationary Time Series

Non-stationary series can be rendered stationary by appropriate transformations before modelling.

- **Differencing:** For a unit-root process Y_t , the first difference $\Delta Y_t = Y_t - Y_{t-1}$ is typically stationary (difference-stationary, DS).
- **Detrending:** If a deterministic trend is present, regress Y_t on a time trend and use the residuals u_t (trend-stationary, TS).
- **Log-transformation:** Often stabilises variance for series with exponential growth.

20.11 Cointegration

When two or more series are individually integrated of order one, $I(1)$, they may share a common stochastic trend. A linear combination

$$Z_t = \alpha_0 + \alpha_1 Y_{1t} + \alpha_2 Y_{2t} + \cdots + \alpha_k Y_{kt}$$

is **cointegrated** if Z_t is stationary, i.e., $Z_t \sim I(0)$. Cointegration implies a long-run equilibrium relationship among the non-stationary variables.

20.11.1 Engle–Granger Test

The Engle–Granger two-step procedure tests for cointegration:

1. **Step 1 (Cointegrating regression)**: Estimate the long-run relationship by OLS,

$$Y_{1t} = \beta_0 + \beta_2 Y_{2t} + \cdots + \beta_k Y_{kt} + \hat{u}_t.$$

Obtain the residuals $\hat{u}_t$.

2. **Step 2 (Unit-root test on residuals)**: Apply an Augmented Dickey–Fuller (ADF) test to $\hat{u}_t$:

$$\Delta \hat{u}_t = \rho \hat{u}_{t-1} + \sum_{i=1}^p \phi_i \Delta \hat{u}_{t-i} + \varepsilon_t.$$

The null hypothesis $\rho = 0$ (unit root) indicates no cointegration; rejection (significantly negative ρ) provides evidence of cointegration.

Critical values for the residual-based ADF test are different from the standard DF table; they are tabulated in Engle and Granger (1987).

20.11.2 Error-Correction Mechanism (ECM)

If variables are cointegrated, their short-run dynamics can be modelled with an ECM. For a bivariate case:

$$\Delta Y_{1t} = \alpha_0 + \lambda \hat{u}_{t-1} + \sum_{i=1}^p \gamma_i \Delta Y_{1,t-i} + \sum_{j=1}^q \delta_j \Delta Y_{2,t-j} + \varepsilon_t,$$

where $\hat{u}_{t-1}$ is the lagged residual from the cointegrating regression (the error-correction term). The coefficient λ measures the speed of adjustment back to the long-run equilibrium; $\lambda < 0$ indicates that deviations are corrected over time.

The ECM can be extended to multiple variables (Vector Error-Correction Model, VECM) by stacking the differenced equations and including the common error-correction term(s) derived from the cointegrating vectors.

21 Time Series Econometrics: Forecasting

21.1 Granger Causality

Let $\{X_t\}$ and $\{Y_t\}$ be two covariance-stationary series. X_t **Granger-causes** Y_t if past values of X improve the forecast of Y beyond what is achievable using only past values of Y . Formally, estimate the unrestricted VAR(p) model

$$Y_t = \alpha_0 + \sum_{i=1}^p \alpha_{1i} Y_{t-i} + \sum_{i=1}^p \beta_{1i} X_{t-i} + u_{1t},$$

and the restricted model that omits the X lags. An F -test (or Wald test) of $H_0: \beta_{11} = \dots = \beta_{1p} = 0$ provides the Granger-causality decision.

21.2 VAR Models

A Vector Autoregression of order p , VAR(p), stacks k endogenous series $\mathbf{y}_t = (y_{1t}, \dots, y_{kt})'$:

$$\mathbf{y}_t = \mathbf{c} + \sum_{i=1}^p \mathbf{A}_i \mathbf{y}_{t-i} + \mathbf{u}_t,$$

where $\mathbf{c}$ is a $k \times 1$ intercept vector, $\mathbf{A}_i$ are $k \times k$ coefficient matrices, and $\mathbf{u}_t \sim \text{i.i.d.}(0, \Sigma)$. Each equation can be estimated by OLS; the system is identified as long as the lag order p is sufficient to satisfy the order condition.

21.3 AR, MA, ARMA, & ARIMA

- **AR(p)**: $y_t = \phi_0 + \sum_{i=1}^p \phi_i y_{t-i} + \varepsilon_t$.
- **MA(q)**: $y_t = \mu + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \varepsilon_t$.
- **ARMA(p, q)**: combines AR and MA parts, $y_t = \phi_0 + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \varepsilon_t$.
- **ARIMA(p, d, q)**: apply d differences to a non-stationary series, $\Delta^d y_t$ follows an ARMA(p, q) process.

21.4 Box–Jenkins Approach

A systematic procedure for building ARIMA models:

1. **Identification**: use plots of the series, ACF and PACF, and unit-root tests to decide on differencing order d and tentative p, q .
2. **Estimation**: estimate the ARIMA(p, d, q) parameters by maximum likelihood (or conditional least squares).
3. **Diagnostic checking**: examine residual ACF, perform Ljung–Box Q tests, and verify normality/heteroskedasticity.
4. **Forecasting**: generate h -step forecasts using the fitted model; prediction intervals are obtained from the estimated error variance.

21.5 Identification

In the context of ARMA modelling, identification relies on the characteristic patterns of the sample ACF and PACF:

- **AR**(p): PACF cuts off after lag p , ACF tails off.
- **MA**(q): ACF cuts off after lag q , PACF tails off.
- **ARMA**(p, q): both ACF and PACF tail off; trial-and-error or information criteria (AIC, BIC) guide the final choice.

21.6 ARCH

Autoregressive Conditional Heteroskedasticity models the variance of a series as a function of past squared residuals:

$$\varepsilon_t = \sigma_t z_t, \quad z_t \sim \text{i.i.d.}(0, 1),$$
$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^p \alpha_i \varepsilon_{t-i}^2, \quad \alpha_0 > 0, \alpha_i \geq 0.$$

The ARCH(p) model captures volatility clustering often observed in financial returns.

21.7 GARCH

Generalized ARCH adds lagged conditional variances:

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^p \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2, \quad \alpha_0 > 0, \alpha_i, \beta_j \geq 0, \sum_i \alpha_i + \sum_j \beta_j < 1.$$

The GARCH(1,1) specification is the most common; it provides a parsimonious yet flexible description of time-varying volatility and is used for forecasting conditional variance and Value-at-Risk calculations.